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Flying Operations

KC-46 AIRCREW EVALUATION CRITERIA

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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This manual implements Air Force Policy Directive (AFPD) 11-2, *Aircrew Operations* and Air Force Instruction AFI 11-200, *Aircrew Training, Standardization/Evaluation, and General Operations Structure,* and is consistent with AFI 11-202, Volume 2, *Aircrew Standardization and Evaluation Program.* It establishes evaluation criteria for the operation of KC-46 aircraft to accomplish their worldwide mobility missions safely and successfully. The authorities to waive wing or unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See DAFI 33-360, *Publications and Forms Management*, for a description of the authorities associated with the tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, or alternately, to the requestor's commander for non-tiered compliance items. This manual applies to all commanders, operations supervisors, and aircrew assigned or attached to all flying activities of commands operating KC-46 aircraft. It applies to all civilian employees and uniformed members of the Regular Air Force, Air Force Reserve, and Air National Guard (ANG). This publication does not apply to the U.S. Space Force. This publication may be supplemented at any level, but all supplements must be routed to the office of primary responsibility (OPR) of this publication for coordination prior to certification and approval. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction 33-322*, Records Management and Information Governance Program,* and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. This manual requires the collection and or maintenance of information protected by the Privacy Act of 1974 authorized by 37 U.S.C. §§ 301a and 334, DoDI 7730.67, *Aviation Incentive Pays and Bonus Program*. The applicable System of Records Notice(s) F011

AF XO A, *Aviation Resource Management System* is available at: **https://dpcld.defense.gov/privacy/SORNS.aspx**. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*; route AF Form 847s from the field through the appropriate functional's chain of command in 847 Central (**https://cs2.eis.af.mil/sites/12797/sitepages/847%20central.aspx**). Compliance with the attachments in this publication is mandatory.

Chapter 1—GENERAL INFORMATION **4**

1.1. General. 4
1.2. Roles and Responsibilities. 4
1.3. Key Words and Definitions. 4
1.4. Deviations and Waivers. 4
1.5. Supplements and Local Procedures. 5
1.6. Evaluations. 5
1.7. Evaluation Requirements. 5
1.8. Grading Guidance. 7
1.9. Grading System. 8
1.10. Unsatisfactory Performance. 8
1.11. Conduct of Evaluations. 8
1.12. Use of AF Form 3862, Flight Evaluation Worksheet. 8
1.13. Aircrew Testing. 9
1.14. Preplanned Mission Data. 9
1.15. Currency of Flight Publications. 9
1.16. Reporting Deviations/Discrepancies by Unlike Specialty. 10

Chapter 2—PILOT EVALUATIONS **11**

2.1. General. 11
2.2. Instrument Evaluations (Initial, Periodic, and Re-qualification). 11
2.3. Qualification Evaluations (Initial, Periodic, and Re-qualification). 11
2.4. Mission Evaluation. 11
2.5. Senior Staff Evaluations. 12
2.6. Operational Mission Evaluation (OME). 12
2.7. Instructor Evaluation (Initial, Periodic, or Re-qualification). 13
2.8. Emergency Procedures Evaluation (EPE). 13

2.9. Supplemental Information. 13
2.10. Pilot Grading Criteria..... 13
Table 2.1. General Pilot Tolerances..... 17
Table 2.2. Landing Tolerances..... 19
Table 2.3. Instrument Tolerances..... 21
Table 2.4. Precision Approach Tolerances. 23
Table 2.5. Non-Precision Approach Tolerances. 24
Chapter 3—BOOM OPERATOR EVALUATIONS 31
3.1. General..... 31
3.2. Qualification Evaluations..... 31
3.3. Mission Evaluations..... 31
3.4. Instructor Evaluations. 32
3.5. Emergency Procedures Evaluation (EPE)..... 32
3.6. Boom Operator Grading Criteria. 32
3.7. QUALIFICATION 36
3.8. MISSION. 37
3.9. Instructor..... 38
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 40
Attachment 2—FLIGHT EVALUATION WORKSHEET EXAMPLES 45

Chapter 1

GENERAL INFORMATION

1.1. General. This Air Force Manual (AFMAN) provides flight examiners and aircrews with procedures and evaluation criteria/tolerances to be used during flight evaluations as specified in AFI 11-202V2 *Aircrew Standardization and Evaluation Program*. Specific areas for evaluation are prescribed to ensure an accurate assessment of the proficiency and capabilities of aircrews. Examiners will use this AFMAN when conducting aircrew evaluations. **(T-2).** Instructors will use this AFMAN when preparing aircrews for qualification. **(T-2).**

1.2. Roles and Responsibilities.

1.2.1. MAJCOM/A3. Air Mobility Command (AMC) is designated lead command for the KC-46 Mission Design Series (MDS) aircraft and is responsible for issuing guidance to establish and standardize aircrew evaluations in coordination with user commands.

1.2.2. Group Commander (OG/CC) or equivalent. OG/CCs are responsible for establishing and maintaining the standardization and evaluation program and ensure examiners administer evaluations in accordance with AFI 11-202V2 and this AFMAN.

1.2.3. Squadron Commander (SQ/CC) or designated representative. SQ/CCs are responsible for establishing and maintaining the squadron standardization and evaluation program and ensuring examiners administer evaluations in accordance with AFI 11-202V2 and this AFMAN.

1.2.4. Flight Examiners. Flight examiners are responsible for administering evaluations in accordance with AFI 11-202V2 and AFMAN 11-2KC-46V2.

1.3. Key Words and Definitions.

1.3.1. “Will” and “Shall” indicate a mandatory requirement.

1.3.2. “Should” is normally used to indicate a preferred, but not mandatory, method of accomplishment.

1.3.3. “May” indicates an acceptable or suggested means of accomplishment.

1.3.4. “Note” indicates operating procedures, techniques, etc., which are considered essential to emphasize.

1.4. Deviations and Waivers. Do not deviate from guidance in this AFMAN except to protect life, preserve safety of flight, or when an in-flight emergency requires immediate action. Directive guidance (will, shall, must, etc.) throughout this regulation is tiered in accordance with DAFI 33-360. For examples of tiered waivers, see DAFI 33-360. MAJCOM/A3 is waiver authority for MAJCOM supplements to this manual. MAJCOM/A3s will forward a copy of approved long-term waivers to this manual to AMC/A3VK for follow-on action.

1.4.1. Deviations. The Pilot in Command (PIC) shall report deviations or exceptions taken without a waiver through command channels to their Chief, Major Command (MAJCOM) Stan/Eval, who in turn shall notify Chief, AMC Stan/Eval as appropriate for follow-on action. **(T-2).**

1.4.2. For the purposes of this manual, flying MAJCOMS are: ACC (Air Combat Command), AETC (Air Education and Training Command), AFGSC (Air Force Global Strike Command), AFMC (Air Force material Command), AFRC (Air Force reserve Command), AFSOC (Air Force Special Operations Command), AMC (Air Mobility Command), NGB (National Guard Bureau), PACAF (Pacific Air Forces), and USAFE-AFAFRICA (United States Air Forces Europe-Air Forces Africa). Commander Air Force forces (COMAFFORs) in the grade of O-8 or higher in Combatant Commands (CCMDs) are considered MAJCOM commanders only for forces under their operational control.

1.5. Supplements and Local Procedures. This AFMAN is a basic directive. Each user MAJCOM may supplement this AFMAN according to AFI 11-200. MAJCOMs may specify unique evaluation items in their appropriate supplement. Units may supplement this AFMAN. Supplements and local procedures cannot be less restrictive than the provisions of this AFMAN or the appropriate flight manual.

1.5.1. **Supplement Coordination Process.** Forward MAJCOM/A3 approved supplements, with attached AF Form 673, *Air Force Publication/Form Action Request***,** to lead command (AMC/A3) for review. AMC/A3 will provide a recommendation and forward to AF/A3T for approval (in accordance with AFI 11-200). Use the following OPR's address: AMC/A3V, 402 Scott Dr., Unit 3A1, Scott AFB IL, 62225-5302. When supplements are published, units will send a final copy to AMC/A3V. **(T-1).**

1.5.2. If necessary, request and include approved long-term waivers to this AFMAN (including, approval authority, date, and expiration date) in the appropriate MAJCOM supplement.

1.5.3. Local Procedures Coordination. Units send a copy of the local supplement to parent MAJCOM/A3V for coordination and approval. When local supplements are published, notify or send a final copy to AMC/A3V, parent MAJCOM, and appropriate NAF, if applicable.

1.6. Evaluations. This manual establishes standardized instrument, qualification, mission, and instructor evaluation criteria. It also establishes the areas/subareas necessary for the successful completion of evaluations, and which required areas/subareas will be considered critical and/or non-critical.

1.7. Evaluation Requirements. Accomplish evaluations concurrently, whenever practical. Flight examiners will evaluate crew resource management (CRM) and threat and error management (TEM) skills on all evaluations. **(T-2).** KC-46 aircrew members will complete the following evaluations at 17-month frequency according to AFI 11-202V2, and the appropriate MAJCOM supplement:

1.7.1. The KC-46 weapon system trainer (WST) and boom operator trainer (BOT) will be used in conjunction with all emergency procedures evaluations (EPEs), instrument (INSTM), qualification (QUAL), and if applicable, mission (MSN) evaluations. **(T-2).** Evaluations will only be performed in approved simulators and must be conducted by an Air Force flight examiner (not contractors). **(T-2).** WST evaluations will only consist of areas/subareas that can be realistically accomplished and are aircrew training device (ATD)-creditable per AFMAN 11-2KC-46V1, *KC-46A Aircrew Training*. **(T-2).** Conduct a thorough pre-mission briefing and post-mission debriefing to the examinee and applicable aircrew members on all aspects of the evaluation. At a minimum, on all evaluations where the examinee is downgraded

in CRM/TEM, examiners will use the AMC Form 4031, *CRM/TEM Skills Criteria Training/Evaluation*, or other MAJCOM-approved assessment form(s) to critically document and evaluate crew resource management and threat and error management (CRM/TEM) skills. **(T-2).** This form will be used in feedback during evaluation debriefs and submitted to AMC Training (AMC/A3T) for CRM trend analysis in accordance with AFI 11-290_AMCSUP, *Cockpit/Crew Resource Management Program.*

1.7.2. Instrument (INSTM) Evaluation. All KC-46 pilots will successfully complete a periodic instrument valuation including the requisite instrument refresher course (IRC) open-book written examination according to AFMAN 11-210, *Instrument Refresher Program (IRP),* and a simulator evaluation. **(T-2).**

1.7.3. Qualification (QUAL) Evaluation. All KC-46 aircrew members will successfully complete a periodic qualification evaluation including the requisite open-book, closed-book examinations, memory action item evaluation, emergency procedures evaluation (EPE), and an ATD/flight evaluation. **(T-2).**

1.7.4. Mission (MSN) Evaluations. All KC-46 mission-ready crew members will complete a mission evaluation. **(T-2).** Whether in the aircraft or ATD, mission evaluations will be flown in accordance with current tactics, unit taskings, and theater area of responsibility scenarios. **(T-2).** Scenarios that represent unit designated operational capability (DOC) tasking and current unit requirements satisfy the intent of this evaluation.

1.7.5. Instructor (INSTR) Evaluations. To initially qualify as an instructor in the KC-46, crew members will successfully complete an initial instructor qualification course and evaluation. **(T-2).** Subsequently, aircrew members designated as instructors will be evaluated on their ability to instruct during all periodic evaluations. **(T-2).** Crewmembers may use the initial instructor evaluation to satisfy the requirements of the periodic QUAL and MSN evaluations. Refer to the specific aircrew chapter for requirements.

1.7.6. SPOT Evaluations. A SPOT evaluation is an evaluation not intended to satisfy the requirements of a periodic (i.e., INSTM, QUAL, MSN, or INSTR) evaluation. SPOT evaluations have no specific requisites or requirements unless specified in MAJCOM supplements or this AFMAN. See AFI 11-202V2 for options available to convert a SPOT evaluation to meet requirements of a periodic evaluation.

1.7.7. Emergency Procedures Evaluation (EPE). See AFI 11-202V2. SPOT evaluations do not require an EPE. Accomplish the EPE in conjunction with QUAL/INST/MSN evaluations as described in chapters **2 and 3** of this publication.

1.7.7.1. Units may develop and maintain a list of EPE program requirements (topics, special interest, etc.) in local supplement or OI. The EPE should include areas commensurate with the examinee's graduated training (e.g. initial, line, instructor, examiner), or as specified in AFI 11-202V2, and MAJCOM supplement.

1.7.7.2. Examinees may use publications that are normally available in-flight. The examinee must demonstrate all memory items applicable to their crew position listed in the FCOM and provide the initial steps of selected non-normal procedures that include memory items that would not allow time for referencing the quick reference handbook (QRH). **(T-2).**

1.7.7.3. Examinees receiving an overall EPE grade of unqualified will be placed in supervised status until recommended additional training and re-evaluation are completed. **(T-2).** Examinees receiving an overall EPE grade of unqualified because of unsatisfactory memory item procedures will not be permitted to fly in their aircrew position until a successful re-evaluation is accomplished. **(T-2).** Accomplish additional training in accordance with AFI 11-202V2. **(T-2).**

1.7.8. Evaluation Prefixes. Use AFI 11-202V2 evaluation prefixes for AF Form 8, *Certificate of Aircrew Qualification*, and AF Form 942, *Record of Evaluation*.

1.7.8.1. For a re-qualification (RQ) evaluation that results from a failed periodic or SPOT evaluation, include all areas under GENERAL and any subarea graded "U". **Note:** Evaluations intended to re-qualify a crewmember after a failed SPOT evaluation will be documented as "RQ SPOT" unless the examination is intended to re-align a periodic evaluation.

1.7.8.2. Conduct RQ evaluations for an expired qualification due to loss of currency, expired periodic evaluation, or commander-directed downgrade in accordance with AFI 11-202V2.

1.7.8.3. Difference Evaluation. The phrase "difference" is used to describe the evaluation of one or more areas to meet qualification requirements. Normally, a difference evaluation will include areas that are different between aircraft models, systems, or operations not previously qualified to operate. A difference evaluation does not have an expiration date established because the evaluation does not satisfy the requirements for the "full" periodic evaluation. For administrative purposes, annotate AF Form 8, flight phase as a SPOT evaluation (according to AFI 11-202V2) and **paragraph 1.7.6** above.

1.8. Grading Guidance.

1.8.1. The overall qualification level awarded for an evaluation is based on performance during both the flight and ground phases. This grade should be awarded only after all evaluation requirements have been completed and given due consideration.

1.8.2. To receive a qualified grade on an evaluation, the aircrew member must satisfy the criteria set forth for that evaluation and demonstrate the ability to operate the aircraft and/or equipment safely and effectively during all phases of the evaluation.

1.8.3. Use the grading criteria in this manual to grade areas/subareas accomplished during an evaluation. The flight examiner may grade any area/subarea accomplished during an evaluation if performance in that area/subarea impacts the specific evaluation accomplished or the safety of flight.

1.8.4. When in-flight or ATD evaluation of a required area is not possible, the area may be verbally evaluated. See AFI 11-202V2_AMCSUP, *Aircrew Standardization and Evaluation Program* and the appropriate chapter of this publication for areas prohibited from verbal/ATD evaluation.

1.8.5. Grading criteria tolerances assume smooth air and stable aircraft conditions. Minor momentary deviations are acceptable, provided the examinee applies prompt corrective action and such deviations do not jeopardize flight safety. Consider cumulative deviations when determining the overall grade.

1.8.5.1. For pilots only, if the flight manual recommends a specific airspeed range for performance of a maneuver, the flight examiner will apply the grading criteria to the upper and lower limits of that range. **(T-2).**

1.8.5.2. Flight examiners will use sound judgment in the application of the grading criteria in this manual to determine the final grade. **(T-2).**

1.9. Grading System. Refer to AFI 11-202V2.

1.10. Unsatisfactory Performance. Refer to AFI 11-202V2.

1.11. Conduct of Evaluations.

1.11.1. Flight examiners will pre-brief the examinee on the conduct, purpose, requirements of the evaluation, and all applicable evaluation criteria. **(T-2).** Flight examiners will then evaluate the examinee in each graded area/subarea. **(T-2).** Flight examiners should not evaluate personnel they have primarily trained, recommended for upgrade evaluation, or who render their performance reports.

1.11.2. Unless otherwise specified, flight examiners may conduct the evaluation in any crew position/seat which will best enable the flight examiner to observe the examinee's performance.

1.11.3. Note discrepancies and deviations from prescribed tolerances and performance criteria during the evaluation. Compare the examinee's performance with the tolerances provided in the grading criteria and assign an appropriate grade for each area.

1.11.3.1. An evaluation will not be changed to a training mission to avoid documenting substandard performance, nor will a training mission be changed to an evaluation. **(T-2).**

1.11.3.2. The judgment of the flight examiner, guidance provided in AFI 11-202V2, and this manual will be the determining factors in assigning an overall grade. **(T-2).** The flight examiner will thoroughly critique all aspects of the flight. **(T-2).** During the critique, the flight examiner will review the examinee's overall rating, specific deviations, area/subarea grades assigned, and any additional training required. **(T-2).**

1.11.3.3. In the event of unsatisfactory performance, the flight examiner will recommend additional training requirements. **(T-2).** Normally, additional training should not be accomplished on the same flight. **Exception:** Additional training on the same flight is allowed when, in the examiner's judgment, unique situations presenting valuable training opportunities (e.g., thunderstorm avoidance, crosswind landings) exist. This option requires utmost flight examiner discretion and judicious application. When used, the examinee must be informed of when the additional training begins and ends. **(T-2).**

1.11.4. The Weapon System Trainer/Boom Operator Trainer/Fuselage Trainer (WST/BOT/FuT) may be used to accomplish additional training and re-checks. Areas for additional training and re-checks should be limited to those areas/subareas that can be realistically accomplished in the WST/BOT/FuT.

1.11.5. Re-checks should be administered by a flight examiner other than the one who administered the original evaluation.

1.12. Use of AF Form 3862, *Flight Evaluation Worksheet*. Units (normally Operations Group Standardization and Evaluations (OGV)) will overprint AF Form 3862 (or electronic equivalent),

using the examples in **Attachment 2** as an evaluation worksheet. **(T-2).** Copy each title, area number, and text (in the order illustrated) to the appropriate blocks. Units may add special interest items and/or local evaluation requirements. In-flight, use the worksheet to ensure all required areas are evaluated. Record positive and negative trend information and aircrew member's performance. File the signed AF Form 3862 or signed draft copy of the AF Form 8 in the aircrew member's Flight Evaluation Folder (FEF) in accordance with AFI 11-202V2. Maintain until the completed AF Form 8 is filed in the FEF, and then discard.

1.13. Aircrew Testing. See specific testing requirements in AFI 11-202V2 and appropriate supplements:

1.13.1. Open-Book Exam. The open-book examination should normally be administered before the initial flight evaluation and subsequently with periodic flight evaluations. The open-book examination will consist of 50-100 questions. **(T-2).** The examination questions will come from a secure question bank (SQB) created and managed by each OGV. Open-book exams will be derived from KC-46 Flight Manuals, AFMAN 11-202V3, AFMAN 11-218 *Aircraft Operations and Movement on the Ground*, Allied Tactical Publication (ATP 3.3.4.2), local supplements, flight crew information file/flight crew bulletin (FCIFs/FCB), and others publications at the unit's discretion. **(T-2).** A separate (unique) INSTR open-book examination is not required for periodic evaluations.

1.13.2. Initial Instructor Open-Book Exam. Administer an initial instructor open-book one time before the initial instructor flight evaluation. The instructor open-book examination is requisite for INIT INSTR flight evaluations only. The examination will have a minimum of 20 questions from AFMAN 11-2KC-46 V1,V2, and V3, (including MAJCOM supplements) and other common flight or instructor-related sources. Questions should include scenario-driven instructor related questions.

1.13.3. Closed-Book Exam. The closed-book examination should normally be administered before the initial flight evaluation and subsequently with periodic flight evaluations. The closed-book exam will be constructed in accordance with AFI 11-202V2, para 6.4.2. **(T-2).**

1.13.4. Instrument Exam. See AFI 11-202V2 requirements.

1.14. Preplanned Mission Data. When using a mission planning agency-derived flight plan, fuel log, etc., do not grade the examinee for those portions computed by the mission planners. Initial data input is the responsibility of the individual developing the mission planning products. However, the examinee is responsible to ensure the data used is accurate and effective for mission accomplishment.

1.14.1. Data Transfer Device (DTD). During mission planning, it is the examinee's responsibility to load all applicable mission data files onto the DTD. This can be confirmed by the examiner during mission planning or during DTD loading onto the aircraft.

1.15. Currency of Flight Publications. All issued technical orders, checklists, and publications required for the flight will be checked for currency and accuracy. **(T-2).**

1.15.1. When unit-maintained publications are used, the examinee will ensure currency and suitability. **(T-2).** The unit supplement to AFI 11-202 V2 or this publication may specify additional publications to be checked.

1.15.2. Electronic Flight Bag (EFB). The examinee is responsible for the currency of all publications, technical orders, checklists and FLIP (if applicable) required for flight. **(T-2).** Examinees are required to ensure that the EFB is configured in accordance with the current EFB concept of employment (CONEMP) and/or MAJCOM baseline configuration. **(T-2).**

1.16. Reporting Deviations/Discrepancies by Unlike Specialty. Examiners must report deviations/discrepancies within their aircraft from established procedures/directives in any area, regardless of the individual's crew specialty, to the OGV for evaluation and necessary action. **(T-2).** If the deviation/discrepancy is found to warrant overall Qualification Level 3 AF Form 8 documentation, annotate SPOT in the Flight Phase block. The Chief of OGV will sign as the flight examiner. **(T-2).**

1.16.1. For formation purposes, examiners of like or unlike specialty in other aircraft within the formation who observe deviations/discrepancies affecting the overall safe conduct of the formation will report such deviation/discrepancy to OGV for evaluation and necessary action.

Chapter 2

PILOT EVALUATIONS

2.1. General. This chapter standardizes initial, periodic, and re-qualification evaluations, including the requirements for instrument, qualification, mission, and instructor evaluations. The examinee must satisfactorily demonstrate the ability to perform all pilot duties safely and effectively, including the operation of appropriate aircraft systems, in accordance with applicable flight manuals and Air Force Instructions/Manuals. **(T-2).**

2.2. Instrument Evaluations (Initial, Periodic, and Re-qualification). KC-46 instrument evaluations should be combined with a qualification evaluation (INST/QUAL). Pilots will be evaluated in all items under GENERAL and INSTRUMENT. **(T-2).** Instrument evaluations will be conducted in a certified WST, during an aircraft sortie, or through a combination of both. **(T-2).**

2.2.1. As a minimum, all pilots must demonstrate at least one precision approach and two non-precision approaches. **(T-2).** These items may not be verbally evaluated. **(T-2).**

2.2.2. As a minimum, evaluate a circling approach and CAT II/III ILS approach for initial qualification, if certified. IAN procedures must be evaluated on at least one approach.

2.3. Qualification Evaluations (Initial, Periodic, and Re-qualification). KC-46 qualification evaluations should be combined with the instrument evaluation (INSTM/QUAL). Include all areas under GENERAL and QUALIFICATION. Qualification evaluations will be conducted in a certified WST, during an aircraft sortie, or a combination of both. **(T-2).**

2.3.1. Evaluate MPs on at least one landing in both the left and right seats. Evaluate FPs in the right seat only. Use a contractor-developed scenario or a unit Standardization/Evaluation-approved and flight examiner-provided scenario. Unit/flight examiner-provided scenarios must be coordinated with the contractor a minimum of one day before the evaluation to ensure compatibility with ATD software. **(T-2).**

2.3.2. Evaluate all pilots in rejected takeoff procedures, engine failure after V1, engine out approach and missed approach, and a sampling of other non-normal and memory items. **(T-2).**

2.3.3. PIQ students are not expected to fulfill the role of an aircraft commander on their initial qualification evaluation. Document this evaluation as “FP” in the crew position on the AF Form 8.

2.3.4. FPs selected for PCO will complete an INIT INSTM/QUAL/MSN evaluation focusing on aircraft commander roles and responsibilities. Examinees will primarily occupy the left seat and include receiver AR requirements. **(T-2)**. Document this evaluation as “MP” in the crew position.

2.3.5. Evaluate ACIQ students in the role of aircraft commander during their initial evaluation.

2.4. Mission Evaluation. Evaluate all areas under GENERAL and MSN. The entirety or a portion of the mission evaluation may be conducted in a certified WST except for areas that are not creditable in the WST in accordance with AFMAN 11-2KC-46 V1. Tactics procedures may be conducted in a certified WST or verbally evaluated.

2.4.1. Units will develop employment scenario's consistent with unit taskings and unit designated operational capability (DOC) taskings and current tactics. **(T-2).** At a minimum include an intelligence scenario, commander's intent, mission objective, and unit or HHQ-defined SPINS. The scenario need not encompass the entire sortie.

2.4.2. Tanker Air Refueling (AR). Pilots will be evaluated on Tanker AR and breakaway procedures. **(T-2).** Rendezvous may be evaluated as pilot flying (PF) or pilot monitoring (PM). Conduct a portion of the evaluation with the autopilot off. Pilots will be graded based on procedures applicable to their crew position. **(T-2)**. Breakaway procedures will be evaluated in the aircraft or in a certified WST as PF or PM. **(T-2)**.

2.4.3. Receiver AR. Rendezvous or closure should be evaluated from a minimum of 100 feet of separation. MPs/IPs will demonstrate the ability to hold sustained contact. FPs may be evaluated on rendezvous and overrun procedures, if observed. Receiver breakaway will be evaluated. **(T-2).** Evaluate a receiver breakaway from within the Air to Air Refuling (AAR) envelope. **(T-2).**

2.4.4. When not intended to lead to AC Certification, Pilot Initial Qualification (PIQ) students will not be evaluated in receiver AR. Document crew position for these evaluations as "FP" on the AF Form 8. With the recommendation of the SQ/DO, subsequent periodic evaluations may be intended to lead to AC Certification. These evaluations will include receiver AR in the left seat and will be documented with "MP" for crew position. **(T-2).** All pilots must receive an aircraft commander evaluation (document as "MP") prior to AC Certification. **(T-2)**.

2.5. Senior Staff Evaluations. All senior staff officer initial, periodic, and re-qualification evaluations include (as a minimum) all areas under GENERAL, INSTRUMENT, and QUALIFICATION, except area 20, Engine Out Operations, and area 23, Other Non-normal Procedures.

2.5.1. Evaluate senior staff officers in a random selection of memory items and rejected takeoff procedures. **Note*:*** Since senior staff officers do not maintain a MISSION qualification in these areas, they may not occupy a pilot's seat during Tanker AAR or Receiver AAR with passengers aboard. Annotate the following restriction on AF Form 8, Examiner's Remarks: "RESTRICTION: This evaluation permits operating the aircraft with a qualified KC-46 instructor pilot at all times and must be directly supervised during critical phases of flight."

2.5.2. Time and training conditions permitting, Senior Officers may elect to be trained in accordance with AFMAN 11-2KC-46V1 and evaluated in Tanker AAR and Receiver AAR. If so, no restriction is required, but clearly document that tanker and/or receiver AAR was accomplished in the examiner's remarks.

2.6. Operational Mission Evaluation (OME). All pilots, except those previously certified as Aircraft Commanders in KC-10 or KC-135 aircraft, will complete a one-time operational mission evaluation demonstrating their ability to operate in command of an aircraft performing the unit's primary mission prior to aircraft commander certification. **(T-2).** The following may be used to satisfy this requirement:

2.6.1. A combination INSTM/QUAL/MSN, INIT MSN, or SPOT evaluation on a local training or operational mission may be accomplished provided the following conditions are met:

2.6.1.1. Approved by the OG/CC or equivalent. Blanket approvals will be documented in the unit supplement. **(T-2).**

2.6.1.2. The evaluation is focused on decision making and CRM/TEM.

2.6.2. A SPOT evaluation given on a 618 AOC (TACC)-tasked or locally assigned Dual Role or JA/ATT mission.

2.6.3. In all cases document the OME on the AF Form 8 with the following comment in the remarks section: "This OME was conducted in conjunction with aircraft commander certification."

2.7. Instructor Evaluation (Initial, Periodic, or Re-qualification). Flight examiners will place particular emphasis on the examinee's ability to recognize student difficulties and provide timely, effective, corrective action. **(T-2).** Instructors should capitalize on all opportunities to instruct, regardless of phase of flight.

2.7.1. Initial: Include all areas under GENERAL and INSTRUCTOR. Pilots, who desire to realign their INSTM/QUAL/MSN evaluation during the initial instructor evaluation must also demonstrate all required areas/subareas in INSTRUMENT, QUALIFICATION, and MISSION and complete all required requisite written examinations. **(T-2).** Evaluation will comply with AFI 11-202V2. **(T-2).** For initial/re-qualification IP evaluations, a boom envelope limits demonstration will be performed. **(T-2).** Conduct a portion of the evaluation with tanker autopilot disengaged.

2.7.2. Air refueling boom envelope limit demonstration. During Receiver AAR, limited inadvertent disconnects are permissible during a boom limits demonstration and should not detract from the examinees' performance. The boom limit demonstration is desired on recurring instructor evaluations, but visual references and techniques may be verbally evaluated by ground instruction at the discretion of the examiner.

2.8. Emergency Procedures Evaluation (EPE). Evaluate a pilot's knowledge of emergency procedures and systems knowledge in the simulator portion of all INSTM/QUAL/MSN evaluations (see **paragraph 1.9.7**).

2.9. Supplemental Information.

2.9.1. Flight examiners may conduct evaluations when scheduled as primary aircrew members.

2.9.2. Instructor and flight examiner pilots receiving periodic evaluations may be evaluated in either seat**.** Left seat taxi operations are not required.

2.9.3. Flight examiners, instructor pilots, and touch-and-go certified aircraft commanders will be evaluated during the completion of a touch-and-go as either the pilot flying or the pilot monitoring. **(T-2).**

2.10. Pilot Grading Criteria.

2.10.1. **General.**

2.10.1.1. **Area 1, Directives and Publications.**

2.10.1.1.1. **Q.** Possesses a high level of knowledge of all applicable aircraft directives and publications and understood how to apply both to enhance mission

accomplishment. Required publications (paper or electronic) were current and properly posted. EFB was in proper configuration in accordance with MAJCOM directives.

2.10.1.1.2. **Q-.** Unsure of some directives but could locate information in appropriate publications. Required publications (paper or electronic) were current but improperly posted. Required applications were up to date. EFB configuration was not in accordance with MAJCOM directives.

2.10.1.1.3. **U.** Unaware of established directives and/or could not locate them in the appropriate publication in a timely manner. Required publications (paper or electronic) were not current. EFB configuration was not in accordance with MAJCOM directives and/or data was not current.

2.10.1.2. **Area 2, Mission Preparation/Planning/Performance.**

2.10.1.2.1. **Q.** Checked all factors applicable to flight such as: weather, NOTAMS, alternate airfields, airfield suitability, fuel requirements, charts, etc. Displayed a high level of knowledge of performance capabilities and operating data. Evaluated data intended for use during takeoff/landing after final adjustments and corrections were made:

2.10.1.2.1.1. V1, V2, flap retract speeds: +/-3 KIAS.

2.10.1.2.1.2. Landing speeds: +/-3 KIAS.

2.10.1.2.2. **Q-.** Made minor errors or omissions in checking all factors that could have detracted from mission effectiveness. Marginal knowledge of performance capabilities and/or operating data. Performance calculations exceeded Q limits but did not exceed:

2.10.1.2.2.1. V1, V2, flap retract speeds: +/-5 KIAS.

2.10.1.2.2.2. Landing speeds: +/-5 KIAS.

2.10.1.2.3. **U.** Made major errors or omissions which would have prevented a safe or effective mission. Unsatisfactory knowledge of performance capabilities and/or operating data. Performance calculations exceeded Q- limits.

2.10.1.3. **Area 3, Use of Checklists/Procedures.**

2.10.1.3.1. **Q.** Consistently opened and closed checklists over interphone. Appropriately and timely called for the correct checklist and gave correct responses throughout the mission. Flight manual procedures were applied appropriately.

2.10.1.3.2. **Q-.** Opened and closed checklists over interphone with minor deviations or untimely. Crewmember required prompting for correct response.

2.10.1.3.3. **U.** Did not open and close checklists over interphone. Used or called for incorrect checklist or consistently omitted checklist items. Unable to identify the correct checklist to use for a given situation. Did not complete checklist prior to event. Flight manual procedures were not applied or applied incorrectly.

2.10.1.4. **Area 4, Safety Consciousness (Critical).**

2.10.1.4.1. **Q.** Aware of and complied with all safety factors required for safe aircraft operation and mission accomplishment.

2.10.1.4.2. **U.** Not aware of or did not comply with all safety factors required for safe aircraft operation or mission accomplishment. Attempted to operate aircraft in a dangerous manner.

2.10.1.5. **Area 5, Judgment/Compliance (Critical).**

2.10.1.5.1. **Q.** Prepared and completed mission in compliance with existing regulations and directives. Demonstrated knowledge of operating procedures and restrictions and where to find them in the correct publications.

2.10.1.5.2. **U.** Unaware of established procedures and/or could not locate them in the appropriate publication in a timely manner. Failed to comply with a procedure that could have jeopardized safety or mission success.

2.10.1.6. **Area 6, Crew Coordination/CRM/TEM.** Use AFI 11-290, *Cockpit/Crew Resource Management Program*, applicable MAJCOM supplements, and AF Form 4031, *CRM Skills Criteria Training/Evaluation Form*, as a reference.

2.10.1.6.1. **Q.** Proactively applied appropriate, established CRM/TEM, and flight path management (FPM) concepts throughout the flight/mission. Ensured safe/effective mission accomplishment by anticipating, recognizing, and mitigating relevant threats. Identified and mitigated own and other crewmembers' errors via the proper use of monitoring/crosschecking procedures and through the employment of applicable, verbalize verify monitor (VVM) practices/procedures.

2.10.1.6.2. **Q-.** Reactively and inconsistently, or inadequately applied appropriate, established CRM/TEM and flight path management (FPM) concepts, but did not allow those deficiencies to detract from mission accomplishment and/or flight safety. Unreliably and/or inadequately anticipated, identified, or mitigated relevant threats and/or own or other crewmembers' inconsequential errors.

2.10.1.6.3. **U.** Did not apply appropriate, established CRM/TEM and flight path management (FPM) concepts to ensure safe/effective mission accomplishment. Failed to anticipate, identify, or mitigate relevant threats and/or own or other crewmembers' consequential errors.

2.10.1.7. **Area 7, Communication Procedures.**

2.10.1.7.1. **Q.** Complete knowledge of and compliance with correct communications procedures. Communications (both inter- and intra-cockpit) were used with concise, accurate, and with proper terminology to effectively direct maneuvers or describe the tactical situation. Complied with and acknowledged all required instructions. Familiar with and correctly operated, HAVE QUICK, identification friend or foe (IFF), MDN, datalink, and secure voice equipment, if observed. Applied proper EMCON procedures as directed by the employment scenario.

2.10.1.7.2. **Q-.** Occasional deviations from procedures that required re-transmissions or resetting codes. Slow in initiating or missed several required radio calls. Transmissions contained extraneous matter, were not in proper sequence, or used non-standard terminology. Displayed limited knowledge of HAVE QUICK, IFF, MDN,

datalink, and secure voice equipment, if observed. Extraneous comments over radio/interphone presented minor distractions. Applied EMCON procedures with minor deviations and/or errors.

2.10.1.7.3. **U.** Incorrect procedures or poor performance caused confusion and jeopardized mission accomplishment. Omitted numerous radio calls. Displayed inadequate knowledge of or inability to operate HAVE QUICK, IFF, MDN, datalink, and secure voice equipment. Inaccurate or confusing terminology significantly detracted from situational awareness or mission accomplishment. Applied EMCON procedures with major deviations and/or errors.

2.10.1.8. **Area 8, Aircrew Flight Equipment Systems/Egress.**

2.10.1.8.1. **Q.** Displayed thorough knowledge of location and use of aircrew flight equipment systems. Demonstrated and emphasized the proper operating procedures used to operate aircraft egress devices such as doors, windows, slide rafts, and escape ropes.

2.10.1.8.2. **Q-.** Limited knowledge of location and use of aircrew flight equipment systems. Unsure of the proper operating procedures used to operate some of the aircraft egress devices.

2.10.1.8.3. **U.** Displayed unsatisfactory knowledge of location and use of aircrew flight equipment systems. Unable to properly operate aircraft egress devices or egress the aircraft.

2.10.1.9. **Area 9, Knowledge/Completion of Forms.**

2.10.1.9.1. **Q.** All required forms and/or flight plans were complete, accurate, readable, accomplished on time, and in accordance with applicable directives/procedures in Aircrew e-Pubs and FAA/ICAO directives. Related an accurate debrief of significant events to applicable agencies (intelligence, maintenance, etc.).

2.10.1.9.2. **Q-.** Minor errors on forms and/or flight plans did not affect conduct of the mission. Incorrectly or incompletely reported some information due to minor errors, omissions, and/or deviations.

2.10.1.9.3. **U.** Did not accomplish required forms and/or flight plans. Omitted or incorrectly reported significant information due to major errors, omissions, and/or deviations.

2.10.1.10. **Area 10, Airmanship/Situational Awareness (Critical).**

2.10.1.10.1. **Q.** Executed the assigned mission in a timely, efficient manner. Conducted the flight with a sense of understanding and comprehension.

2.10.1.10.2. **U.** Decisions, or lack thereof, resulted in failure to accomplish the assigned mission.

2.10.1.11. **Area 11, Aircrew Discipline (Critical).**

2.10.1.11.1. **Q.** Provided required direction/information. Correctly adapted to meet new situational demands. Demonstrated strict professional flight and crew discipline throughout all phases of the mission.

2.10.1.11.2. **U.** Failed to provide direction/information when needed. Failed to correctly adapt to meet new situational demands. Failed to exhibit strict flight or crew discipline. Violated or ignored rules or regulations.

2.10.2. **Qualification.** Use the criteria in **Table 2.1** as general tolerances for airspeed, altitude, and heading/course unless specifically noted in another subarea.

Table 2.1. General Pilot Tolerances.

Q	Altitude	+/- 100 feet
	Airspeed	+ 10 / -5 kts
	Heading/Course	+/- 5 degrees
Q-	Altitude	+/- 200 feet
	Airspeed	+ 15 / -10 kts
	Heading/Course	+/- 10 degrees
U		Exceeds Q- Criteria.
NOTES: ***1:*** Airspeed tolerances apply when a specific airspeed has been assigned by Air Traffic Control or prescribed in the flight manual. ***2:*** Add 5 KIAS, 50 feet (when practical), and 2 degrees to all engines operating criteria for operations with an engine out criteria.		

2.10.2.1. **Area 12, Ground Operations/Taxi.**

2.10.2.1.1. **Q.** Established and adhered to station, engine start, taxi, and takeoff time to ensure thorough pre-flight, check of personal equipment, crew/passenger briefings, etc. Accurately determined readiness of aircraft for flight. Completed all systems pre-flight/post-flight inspections in accordance with flight manuals. Conducted taxi operations according to flight manuals, AFMAN 11-218, and local procedures.

2.10.2.1.2. **Q-.** Same as above except for minor procedural deviations that did not detract from mission effectiveness.

2.10.2.1.3. **U.** Crew errors directly contributed to a late takeoff that degraded the mission. Failed to accurately determine readiness for flight. Failed to pre-flight/post-flight a critical component or could not conduct a satisfactory pre-flight/post-flight inspection. Violated flight manual procedures and/or jeopardized safe taxi operations.

2.10.2.2. **Area 13, Takeoff.**

2.10.2.2.1. **Q.** Maintained smooth, positive aircraft control throughout the takeoff. Performed the takeoff in accordance with flight manuals and as published/directed.

2.10.2.2.2. **Q-.** Minor deviations from published procedures without affecting safety of flight. Control was rough or erratic. Hesitant in application of procedures/corrections.

2.10.2.2.3. **U.** Takeoff was potentially dangerous. Exceeded aircraft/systems limitations. Failed to establish proper climb attitude. Excessive deviation from intended flight path.

2.10.2.3. **Area 14, Radar Operations/Weather Avoidance/Windshear.**

2.10.2.3.1. **Q.** Effectively demonstrated procedures for operating weather radar. Updated weather radar/analysis throughout the mission. Highly knowledgeable of windshear detection and avoidance equipment. Used all available sources to determine if and/or to what degree severe weather conditions exist. Complied with all weather separation and windshear avoidance procedures.

2.10.2.3.2. **Q-.** Minor deviations observed when operating weather radar. Did not update radar/weather analysis during worsening weather conditions. Limited knowledge of windshear detection and avoidance equipment.

2.10.2.3.3. **U.** Unable to demonstrate proper use of weather radar. Failed to update radar/weather analysis during the mission. Displayed unsatisfactory knowledge of windshear detection and avoidance equipment. Failed to comply with weather separation or windshear avoidance directives that could have jeopardized safety or mission success.

2.10.2.4. **Area 15, Fuel Conservation.**

2.10.2.4.1. **Q.** Possessed a high level of knowledge of all applicable aircraft publications and directives, understood how to apply both to enhance fuel conservation and fuel planning. Fully understood dispatch flight management and ETOPS procedures (if applicable). Successfully applied fuel conservation procedures in all areas of the mission with less than 10% unidentified extra.

2.10.2.4.2. **Q-.** Possessed some knowledge of applicable aircraft publications and directives and understood how to apply both to enhance fuel conservation and fuel planning. Some understanding of dispatch flight management procedures and CAT I and ETOPS procedures (if applicable). Successfully applied some fuel conservation procedures, but failed to apply fuel conservation procedures in all areas of the mission and planned with more than 10% unidentified extra.

2.10.2.4.3. **U.** Unaware of fuel conservation procedures. Unable to fuel plan. Failed to apply any fuel conservation procedures in the mission.

2.10.2.5. **Area 16, VFR Pattern.**

2.10.2.5.1. **Q.** Performed traffic pattern, turn-to-final, or final approach in accordance with published procedures. Aircraft control was smooth and positive. Constantly cleared area of intended flight.

2.10.2.5.2. **Q-.** Performed traffic pattern, turn-to-final, or final approach with minor deviations to procedures as published/directed. Aircraft control was safe but not

consistently smooth and positive. Over or under-shot final approach, but was able to intercept normal glide path. Adequately cleared area of intended flight.

2.10.2.5.3. **U.** Did not perform traffic pattern, turn-to-final, or final approach in accordance with published procedures. Displayed erratic aircraft control. Did not clear area of intended flight.

2.10.2.6. **Area 17, Landings.**

2.10.2.6.1. Subarea 17A, Normal Landing.

2.10.2.6.2. Subarea 17B, Engine Out.

2.10.2.6.3. Subarea 17C, Touch and Go.

2.10.2.6.4. Reference **Table 2.2** for grading criteria specific to landings.

2.10.2.6.5. Specific items to evaluate include threshold altitude/airspeed, runway alignment, flare, touchdown, and landing in crab.

Table 2.2. Landing Tolerances.

Q	Performed landings as published/directed in accordance with flight manuals.	
	Airspeed	+10/-5 KIAS with proper wind corrections
	Touchdown Distance	750-3,000 feet
	Centerline	+/- 15 feet left or right
Q-	Performed landings with minor deviation to procedures as published/directed.	
	Airspeed	+15/-10 KIAS with proper wind corrections
	Touchdown Distance	500-750 or 3,000-3,500 feet
	Centerline	+/- 25 feet left or right
U	Landing not performed as published/directed. Exceeded Q- criteria.	
NOTES: ***1:*** Airspeed tolerances apply to computed threshold speed. ***2:*** Add 5 KIAS, 50 feet (when practical), and 2 degrees to all engines operating criteria for operations with an engine out criteria.		

2.10.2.7. **Area 18, Landing Roll/Braking.**

2.10.2.7.1. **Q.** Performed as published/directed in accordance with flight manuals. Braking action and/or autobrake setting was appropriate for the conditions.

2.10.2.7.2. **Q-.** Performed landings with minor deviation to procedures as published/directed. Braking action was unnecessarily delayed or not appropriate for the conditions. Autobrake setting was not appropriate for the conditions.

2.10.2.7.3. **U.** Landing not performed as published/directed. Autobrake usage was unsafe or area exceeded Q- criteria.

2.10.2.8. **Area 19, All Engine Go-Around (GA).** Required in-flight only if a GA or engine out GA was not evaluated in the simulator (not required if Area 21 is accomplished).

2.10.2.8.1. **Q.** Initiated and performed GA promptly and according to flight manual and directives. Applied smooth control inputs. Acquired and maintained a positive climb.

2.10.2.8.2. **Q-.** Slow or hesitant to initiate GA. Slightly over-controlled the aircraft. Minor deviations did not affect mission accomplishment or compromise safety.

2.10.2.8.3. **U.** Did not initiate GA when appropriate or directed. Major deviations or misapplication of procedures could have led to an unsafe condition.

2.10.2.9. **Area 20, Engine Out Operations.** Use approach criteria for the type of approach being flown and the following:

2.10.2.9.1. **Q.** Proper control inputs were used to correct asymmetric condition. Proper consideration was given to maneuvering the aircraft with regard to the inoperative engine. Maintained criteria in **Table 2.1**. **(Note 2).**

2.10.2.9.2. **Q-.** Minor deviations in aircraft control allowed the aircraft to occasionally fly uncoordinated flight. Momentarily deviated from criteria in **Table 2.1**. **(Note 2).**

2.10.2.9.3. **U.** Aircraft was not properly trimmed and/or aircraft control was erratic and consistently resulted in uncoordinated flight. Maneuvering the aircraft with regard to the engine out condition was potentially unsafe. Exceeded Q- criteria in **Table 2.1**. **(Note 2).**

2.10.2.10. **Area 21, Engine Out GA/Engine Fail Takeoff Continued.**

2.10.2.10.1. **Q.** Performed all required procedures in accordance with the flight manual and directives. Applied smooth, positive, and coordinated control inputs. Rudder and aileron inputs were in correct direction.

2.10.2.10.2. **Q-.** Procedural errors were made which did not affect safety. Aircraft control was not consistently smooth and positive. Rudder and aileron inputs were in correct direction but some over/under-control.

2.10.2.10.3. **U.** Procedural errors were made which affected safety. Rudder and/or aileron inputs were incorrect.

2.10.2.11. **Area 22, Memory Item Procedures (Critical).**

2.10.2.11.1. **Q.** Correct, timely responses. Maintained aircraft control. Coordinated proper crew actions.

2.10.2.11.2. **U.** Incorrect sequence, unsatisfactory response, or unsatisfactory performance of corrective actions.

2.10.2.12. **Area 23, Other Observed Non-Normal Procedures.**

2.10.2.12.1. **Q.** Operated within prescribed limits and correctly diagnosed problems. Performed/explained proper corrective action for each type of malfunction. Effectively used available aircrew aids and checklists.

2.10.2.12.2. **Q-.** Operated within prescribed limits but slow to analyze problems or apply proper corrective actions. Did not effectively use and/or experienced delays, omissions, or deviations in use of checklist and/or available aids.

2.10.2.12.3. **U.** Attempted to exceed limitations. Unable or failed to analyze problem or take proper corrective action. Did not use checklists or available aids effectively.

2.10.2.13. **Area 24, Systems Operations/Knowledge/Limitations.**

2.10.2.13.1. **Q.** Demonstrated appropriate knowledge and proper procedural use of aircraft systems and operations limitations.

2.10.2.13.2. **Q-.** Demonstrated limited knowledge or proper procedural use of aircraft systems and operations limitations.

2.10.2.13.3. **U.** Unable to demonstrate knowledge or proper procedural use of aircraft systems and operations limitations.

2.10.3. **Instrument.** Use the criteria in **Table 2.3** as general tolerances for airspeed, level-off altitude, and heading/course with all engines operating:

Table 2.3. Instrument Tolerances.

Q	Level-off Altitude	+/- 100 feet
	Airspeed	+ 10 / -5 kts
	Heading/Course	+/- 5 degrees
Q-	Level-off Altitude	+/- 200 feet
	Airspeed	+ 15 / -10 kts
	Heading/Course	+/- 10 degrees
U		Exceeds Q- Criteria.

NOTES:

1: Airspeed tolerances apply when a specific airspeed has been assigned by Air Traffic Control or prescribed in the flight manual. Airspeed "minus" tolerances are based on minimum maneuvering speed for aircraft configuration.

2: Add 5 KIAS, 50 feet (when practical), and 2 degrees to all engines operating criteria for operations with an engine out criteria.

2.10.3.1. **Area 25, Instrument Departure/SID.**

2.10.3.1.1. **Q.** Complied with all restrictions or controlling agency instructions. Made all required reports. Applied course/heading corrections promptly. Demonstrated smooth, positive control.

2.10.3.1.2. **Q-.** Minor deviations in navigation occurred during departure. Slow to comply with controlling agency instructions or unsure of reporting requirements. Slow to apply course/heading corrections. Aircraft control was not consistently smooth and positive.

2.10.3.1.3. **U.** Failed to comply with published/directed departure, or controlling agency instructions. Accepted an inaccurate clearance. Aircraft control was erratic.

2.10.3.2. **Area 26, En Route Navigation/FMC Operations.**

2.10.3.2.1. **Q.** Demonstrated capability to navigate using all available means. Used appropriate navigation procedures. Complied with clearance instructions. Aware of position at all times. Remained within the confines of assigned airspace.

2.10.3.2.2. **Q-.** Minor errors in procedures/use of navigation equipment. Slow to comply with clearance instructions. Experienced some difficulty in establishing exact position and course. Slow to adjust for deviations in time and course.

2.10.3.2.3. **U.** Major errors in procedures/use of navigation equipment. Could not establish position. Failed to recognize checkpoints or adjust for deviations in time and course. Did not remain with the confines of assigned airspace.

2.10.3.3. **Area 27, Holding (If Observed).**

2.10.3.3.1. **Q.** Performed entry and holding in accordance with published procedures and directives and:

2.10.3.3.1.1. Timing +/- 15 seconds.

2.10.3.3.1.2. Distance +/- 2 DME or NM.

2.10.3.3.1.3. EFC: +/- 2 minutes (if assigned).

2.10.3.3.2. **Q-.** Performed entry and holding procedures with minor deviations. Exceeded Q criteria but within:

2.10.3.3.2.1. Timing +/- 20 seconds.

2.10.3.3.2.2. Distance +/- 3 DME or NM.

2.10.3.3.3. **U.** Holding was not in accordance with flight manuals, directives, or published procedures. Exceeded Q- criteria.

2.10.3.4. **Area 28, Use of NAVAIDS.**

2.10.3.4.1. **Q.** Ensured NAVAIDS were properly tuned, identified, and monitored.

2.10.3.4.2. **Q-.** Some deviations in tuning, identifying, and monitoring NAVAIDS.

2.10.3.4.3. **U.** Did not ensure NAVAIDS were tuned, identified, and monitored.

Note: The requirement to tune and identify NAVAIDS can be satisfied by confirming that the tuned NAVAID frequency is replaced by the correct alphabetical identifier on the PFD/ND or by aurally identifying the NAVAID.

2.10.3.5. **Area 29, Descent/Arrival.**

2.10.3.5.1. **Q.** Performed descent as directed. Complied with all flight manual, control-issued, or STAR restrictions in a proficient manner. Accomplished all required checks.

2.10.3.5.2. **Q.** Performed descent as directed with minor deviations that did not compromise mission safety. Slow to accomplish required checks.

2.10.3.5.3. **U.** Performed descent with major deviations. Did not accomplish required checks. Erratic corrections. Exceeded flight manual limitations.

2.10.3.6. **Area 30, Precision Approaches.** Includes subareas PAR and ILS/GLS. Use the criteria in **Table 2.4** as general tolerances for airspeed, altitude, heading, glide slope, and azimuth.

Table 2.4. Precision Approach Tolerances.

Q	Airspeed	+ 10 / -5 kts
	Heading (PAR)	+/- 5 degrees
	Azimuth (ILS/GLS)	Within 1 dot
	Glideslope (ILS/GLS)	Within 1 dot
Q-	Airspeed	+ 15 / -5 kts
	Heading (PAR)	+/- 10 degrees
	Azimuth (ILS/GLS)	Within 2 dots
	Glideslope (ILS/GLS)	Within 2 dots high or 1 dot low
U		Exceeds Q- Criteria.
NOTES:		
1: Airspeed tolerances are based on computed approach speed.		
2: Add 5 KIAS, and 2 degrees to all engines operating criteria for operations with an engine out criteria.		

2.10.3.6.1. **Subarea 30A, Precision Approach Radar (PAR) Approach.**

2.10.3.6.1.1. **Q.** Approach was in accordance with flight manuals, directives, or published procedures. Smooth and timely response to controller's instructions. Established initial glide path and maintained with only minor deviations. Complied with decision height. Position would have permitted a safe landing. Elevation did not consistently exceed slightly above or slightly below glide path.

2.10.3.6.1.2. **Q-.** Performed approach with minor deviations. Slow to respond to controller's instructions and make corrections. Improper glide path control. Complied with decision height. Position would have permitted a safe landing. Elevation did not exceed well-above or well-below glide path.

2.10.3.6.1.3. **U.** Approach not in accordance with flight manuals, directives, or published procedures. Erratic corrections. Did not respond to controller's instructions. Did not comply with decision height and/or position would not have permitted a safe landing. Erratic glide path control. Exceeded Q- criteria.

2.10.3.6.2. **Subarea 30B, Instrument Landing System (ILS)/Ground Based Augmentation System (GBAS) Landing System (GLS) Approach.**

2.10.3.6.2.1. **Q.** Approach was in accordance with flight manuals, directives, or published procedures. Smooth and timely corrections to azimuth and glide slope. Complied with decision height. Position would have permitted a safe landing. Maintained glide path with only minor deviations.

2.10.3.6.2.2. **Q-.** Performed approach with minor deviations. Slow to make corrections. Slow to comply with decision height. Position would have permitted a safe landing. Improper glide path control.

2.10.3.6.2.3. **U.** Approach not in accordance with flight manuals, directives, or published procedures. Erratic corrections. Did not comply with decision height and/or position at decision height would not have permitted a safe landing. Exceeded Q- criteria.

2.10.3.6.3. **Subarea 30C, Category II/III Instrument Landing System (ILS) Approach.**

2.10.3.6.3.1. **Q.** Approach was in accordance with applicable directives/procedures in Aircrew e-Pubs and FAA/ICAO directives. Smooth and timely corrections to azimuth and glide slope. Complied with decision height. Position would have permitted a safe landing. Maintained glide path with only minor deviations.

2.10.3.6.3.2. **Q-.** Performed approach with minor deviations. Slow to make corrections. Slow to comply with decision height. Position would have permitted a safe landing. Improper glide path control.

2.10.3.6.3.3. **U.** Approach not in accordance with flight manual, directives, or published procedures. Erratic corrections. Did not comply with decision height and/or position at decision height would not have permitted a safe landing. Exceeded Q- criteria.

2.10.3.7. **Area 31, Non Precision Approach.** Includes subareas NDB, Localizer/VOR, ASR, TACAN, RNAV/GPS/RNP. Use **Table 2.5** criteria as general tolerances for airspeed, altitude at MDA, heading, course, timing, and distance with all engines operating.

Table 2.5. Non-Precision Approach Tolerances.

Q	MDA	+100 / -0 feet
	Airspeed	+ 10 / -5 kts
	Course (NDB, VOR, TACAN)	+/- 5 degrees
	Azimuth (LOC, RNAV)	Within 1 dot
	Timing	Computed/adjusted timing to determine MAP within 20 seconds (when required).
Q-	MDA	+150 / -50 feet
	Airspeed	+ 15 / -5 kts
	Course (NDB, VOR, TACAN)	+/- 10 degrees
	Azimuth (LOC, RNAV)	Within 2 dots
	Timing	Computed/adjusted timing to determine MAP within 30 seconds (when required).
U	Approach not in accordance with published procedures. Maintained steady-state. Flight below the MDA, even though the -50 foot limit was not exceeded. Position would not have permitted a safe landing. Failed to compute or adjust timing to determine MAP (when required). Exceeded Q- criteria.	

NOTES:

1: Airspeed tolerances are based on computed approach speed.

2: Add 5 KIAS, 50 feet (when practical), and 2 degrees to all engines operating criteria for operations with an engine out criteria.

2.10.3.8. **Area 32, Circling Approach (If Observed).**

2.10.3.8.1. **Q.** Properly identified aircraft category for the approach and remained within the lateral limits for that category. Complied with controller's instructions. Attained runway alignment without excessive bank angles. Did not descend from the MDA until in a position to place the aircraft on a normal glide path or execute a normal landing.

2.10.3.8.2. **Q-.** Slow to identify aircraft category for the approach and remained within the lateral limits for that category. Slow to comply with controller's instructions. Attained runway alignment but occasionally required excessive bank angles or maneuvering.

2.10.3.8.3. **U.** Did not properly identify aircraft category or exceeded the lateral limits of circling airspace. Did not comply with controller's instructions. Excessive maneuvering to attain runway alignment was potentially unsafe. Descended from the MDA before the aircraft was in a position for a normal glide path or landing. Exceeded Q- criteria.

2.10.3.9. **Area 33, Missed Approach.**

2.10.3.9.1. **Q.** Executed missed approach in accordance with applicable directives/procedures in Aircrew e-Pubs and FAA/ICAO directives. Complied with controller's instructions. Applied smooth control inputs.

2.10.3.9.2. **Q-.** Executed missed approach with minor deviations to published procedures. Slow to comply with controller's instructions. Slightly over controlled the aircraft.

2.10.3.9.3. **U.** Did not execute missed approach in accordance with flight manuals, directives, or published procedures. Did not comply with controller's instructions. Deviation or misapplications of procedures could have led to an unsafe condition.

2.10.3.10. **Area 34, Automation Management.**

2.10.3.10.1. **Q.** Adhered to and demonstrated appropriate knowledge of published flight path management (FPM) guidance regarding the operation of automated aircraft flight systems, pilot flying (PF)/pilot monitoring (PM) flight automation responsibilities, and verbalize, verify, and monitor (VVM) procedures as they relate to flight automation. Proficiently programmed, reviewed/verified, and operated automated flight systems at suitable levels to enhance situational awareness and/or to reduce pilot workload. Either did not make flight automation errors or quickly identified and mitigated those errors.

2.10.3.10.2. **Q-.** Demonstrated limited knowledge of published FPM guidance for the operation of automated flight systems, PF, PM flight automation responsibilities, and

VVM guidance regarding automated flight systems. Inconsistently or inadequately programmed, reviewed, verified, or operated aircraft automated flight systems at suitable levels to enhance situational awareness and/or to reduce pilot workload. Made, but, did not identify or mitigate inconsequential flight automation errors.

2.10.3.10.3. **U.** Did not follow published FPM guidance for the operation of automated flight systems, causing detriment to mission/flight accomplishment. Did not adequately employ PF, PM, and/or VVM guidance regarding the usage of flight automation systems. Did not adequately program, review, verify, and/or operate automated aircraft systems at suitable levels. Made, but, did not identify or mitigate consequential flight automation errors.

2.10.3.11. **Area 35, Pilot Monitoring (PM).**

2.10.3.11.1. **Q.** Effectively monitored and supported/advised the PF, intervened, when appropriate, if the PF is not adequately controlling the aircraft flight path. Complied with applicable flight policies, procedures, and made required flight callouts. Remained vigilant to identify, communicate, and mitigate events/distractions that may adversely affect flight path management. Monitored energy and flight path performance and was alert for erroneous/conflicting aircraft control and navigational information. Effectively addressed aircraft system failures or unexpected aircraft flight guidance and aircraft system outcomes.

2.10.3.11.2. **Q-.** Did not fully support/advise the PF regarding the aircraft flight path. Slow to intervene if the PF is not adequately controlling the aircraft flight path. Flight policies and procedures were not fully applied and required flight callouts were inconsistent. Flight path/energy management awareness, communication, and/or vigilance is sporadic but did not adversely affect flight safety. Intermittently addressed aircraft system failures or unexpected aircraft flight guidance and aircraft system outcomes.

2.10.3.11.3. **U.** Failed to support/advise the PF regarding the aircraft flight path. Did not intervene if the PF was not adequately controlling the aircraft flight path. Application of flight policies and procedures were insufficient and required callouts were not made. Flight path/energy management awareness, communication, and/or vigilance was insufficient or jeopardized flight safety. Failed to address aircraft system failures or unexpected aircraft flight guidance and aircraft system outcomes.

2.10.4. **Mission (MSN).**

2.10.4.1. **Area 36, Formation (If Observed).** Includes subareas Lead, Departure, Join-up, En-Route, Breakup, and Position Changes.

2.10.4.1.1. **Q.** Performed maneuver consistent with published guidance. Aircraft control was smooth and positive. Constantly cleared area of intended flight.

2.10.4.1.2. **Q-.** Maneuver perfomed in a manner inconsistent with published guidance. Aircraft control was safe but not consistently smooth and positive. Adequately cleared area of intended flight.

2.10.4.1.3. **U.** Failed to comply with published guidance. Displayed erratic aircraft control. Did not clear area of intended flight.

2.10.4.2. **Area 37, Tactics Techniques Procedures (TTP)/Execution.** Includes subareas: Tactical Situational Awareness System (TSAS) and Datalink, Defensive Systems/Tactics and Threat Avoidance.

2.10.4.2.1. **Q.** Applied tactics consistent with the threat, current directives, and good judgement. Executed the plan and achieved mission goals. Quickly adapted to changing environment and maintained situational awareness. Performed maneuvers consistent with published guidance. Aircraft control was smooth and positive. Constantly cleared area of intended flight.

2.10.4.2.2. **Q-.** Applied tactics with only minor deviations, omissions, and/or errors which degraded mission effectiveness but did not prevent the successful accomplishment of the overall mission objective. Maneuvers performed in a manner consistent with published guidance. Aircraft control was safe but not consistently smooth and positive. Adequately cleared area of intended flight.

2.10.4.2.3. **U.** Unable to accomplish the mission due to major errors or omissions during execution of the tactical plan. Situation awareness was lost. Numerous threat reactions were omitted or incorrect. Maneuvers performed in a manner inconsistent with published guidance. Displayed erratic aircraft control. Did not clear area of intended flight.

2.10.4.3. **Area 38, Tanker AAR.** Includes subareas Rendezvous, Platform Control, Breakaway, and Overrun Procedures. **Note:** When refueling with autopilot off, add 100 feet, 5 KIAS, and 5 degrees to all tolerances.

2.10.4.3.1. **Q.** Aircraft control was smooth and positive. Performed all checklists and complied with procedures outline in the flight manual and other governing directives. Met the following criteria:

2.10.4.3.1.1. Airspeed: +10 / -10 KIAS

2.10.4.3.1.2. Altitude: +/- 200 feet

2.10.4.3.2. **Q-.** Aircraft control was not always smooth and positive, but was adequate. Accomplished procedures required by the flight manual, checklists, and other governing directives with deviation/omissions which did not affect safety of flight. Exceeded Q criteria but does not exceed:

2.10.4.3.2.1. Airspeed: +15 / -15 KIAS

2.10.4.3.2.2. Altitude: +/- 300 feet

2.10.4.3.3. **U.** Had deviations/omissions that affected flight safety and/or the successful completion of AAR. Exceeded Q- limits.

2.10.4.4. **Area 39, Receiver AAR.** Includes subareas Rendezvous, Closure, AAR Position/Control, Breakaway, Overrun procedures, and Right seat AAR limit demonstration.

2.10.4.4.1. **Q.** Established and maintained proper refueling position. Aircraft control was positive and smooth. Demonstrated a complete knowledge of rendezvous and closure procedures. Performed all procedures in accordance with applicable checklists

and other governing directives. Met the following criteria: Inadvertent Disconnects: Minimal.

2.10.4.4.2. **Q-.** Slow to recognize and apply needed corrections to establish and maintain proper refueling position. Aircraft control was not always positive and smooth, but was adequate. Accomplished rendezvous and closure with deviations which did not affect safety of flight or the successful completion of AAR. Performed all procedures in accordance with applicable checklists and other governing directives with only minor omissions or deviations. Exceeded Q criteria but did not exceed: Inadvertent Disconnects: Multiple.

2.10.4.4.3. **U.** Erratic or dangerous in the pre-contact/refueling position. An excessive number of inadvertent disconnects resulted. Had deviations/omissions that affected safety of flight or successful completion of AAR. Did not perform all procedures in accordance with applicable checklists and other governing directives or omitted major items. Exceeded Q- limits.

2.10.4.5. **Area 40. Night Vision Goggles (NVG) Operations (If Observed).**

2.10.4.5.1. **Subarea 40A. NVG Usage/Limitations (If Observed).**

2.10.4.5.1.1. **Q.** Properly pre-flighted, handled, and utilized NVGs in accomplishment of the mission. Recognized benefits/limitations of NVGs.

2.10.4.5.1.2. **Q-.** Made minor omissions or deviations during pre-flight or utilization of NVGs. Caused no serious damage to equipment/aircraft and did not negatively affect the mission.

2.10.4.5.1.3. **U.** Used incorrect procedures for NVGs, which caused damage to equipment/aircraft or mission failure.

2.10.4.5.2. **Subarea 40B, NVG Takeoff (If Observed).**

2.10.4.5.2.1. **Q.** Maintained smooth, positive aircraft control throughout the takeoff. Performed the takeoff in accordance with applicable directives/procedures in Aircrew e-Pubs and FAA/ICAO directives. Accomplished rejoin to correct formation position with only momentary deviations.

2.10.4.5.2.2. **Q-.** Minor deviations from published procedures without affecting safety of flight. Control was rough or erratic. Hesitant in application of procedures/corrections. Formation position varied considerably. Slow to correct position.

2.10.4.5.2.3. **U.** Takeoff was potentially dangerous. Exceeded aircraft systems/limitations. Failed to establish proper climb attitude. Excessive deviation from intended flight path. Violated flight manual procedures. Did not maintain safe separation. Exceeded Q- criteria.

2.10.4.5.3. **Subarea 40C, NVG Approach and Landing (If Observed).** If at night, an NVG approach and landing should be accomplished. For landing, use **Table 2.2** for grading criteria.

2.10.4.5.3.1. **Q.** Followed procedures as briefed. Smooth, positive control throughout the approach. Aircraft in position to intercept glide path. Constantly

cleared area of intended flight.

2.10.4.5.3.2. **Q-.** Performed approach with minor deviations to published procedures. Aircraft control was not consistently positive and smooth. Over/under-shot final approach slightly but was able to intercept glide path.

2.10.4.5.3.3. **U.** Approach not performed in accordance with applicable directives/procedures in Aircrew e-Pubs and FAA/ICAO directives. Displayed erratic aircraft control. Over/under-shot final approach or potentially unsafe maneuvering to intercept final. Did not clear area of intended flight.

2.10.4.6. **Area 41, Nuclear Plans/Alert Procedures (If Observed).**

2.10.4.6.1. **Q.** Demonstrated adequate knowledge of plans applicable to the unit mission. Thoroughly familiar with alert procedures and contingencies.

2.10.4.6.2. **Q-.** Demonstrated limited knowledge or comprehension of plans or alert procedures applicable to the unit.

2.10.4.6.3. **U.** Failed to demonstrate sufficient knowledge of plans/alert procedures to ensure effective mission accomplishment.

2.10.5. **Instructor.**

2.10.5.1. **Area 42, Instructor Ability (Critical).**

2.10.5.1.1. **Q.** Demonstrated the ability to communicate effectively. Provided appropriate guidance when necessary. Planned ahead and made timely decisions. Identified and corrected potentially unsafe maneuvers/situations.

2.10.5.1.2. **U.** Unable to effectively communicate or provide timely feedback to the student. Gave instruction that was unsafe or contradicted published directives. Did not provide corrective action when necessary. Did not plan ahead or anticipate student problems. Did not identify an unsafe maneuvers/situations in a timely manner. Made no attempt to instruct.

2.10.5.2. **Subarea 42A, Demonstration of Maneuvers (Critical).**

2.10.5.2.1. **Q.** Effectively demonstrated correct procedures, systems operation, or flight maneuvers. Thorough knowledge of applicable aircraft systems, procedures, publications, and directives.

2.10.5.2.2. **U.** Ineffective or incorrect demonstration of procedures, systems operation, or flight maneuvers. Insufficient depth of knowledge about applicable aircraft systems, procedures, and/or proper source material.

2.10.5.3. **Subarea 42B, Student Briefing/Critique (Critical).**

2.10.5.3.1. **Q.** Briefings were well organized, accurate, and thorough. Reviewed student's present level of training and defined mission events to be performed. During the critique, demonstrated an effective ability to reconstruct the flight, offer mission analysis, and provide guidance, where appropriate. Training grade reflected the actual performance of the student relative to the standard. Pre-briefed the student's next mission, if required.

2.10.5.3.2. **U.** Briefings were marginal or non-existent. Did not review student's past performance. Failed to adequately critique student or analyze the mission. Training grade did not reflect actual performance of student. Overlooked or omitted major discrepancies. Incomplete pre-briefing of student's next mission, if required.

Chapter 3

BOOM OPERATOR EVALUATIONS

3.1. General. This chapter standardizes initial, periodic, and re-qualification evaluations, including the requirements for qualification, mission, and instructor evaluations.

3.2. Qualification Evaluations. Qualification evaluations shall be administered on actual air refueling missions. **(T-2).** A minimum of one boom contact must be accomplished for the qualification evaluation. **(T-2).** Closed book test, open book test, EPE and memory items will be annotated on the AF Form 8. **(T-2).**

3.2.1. Initial Evaluation: Evaluate all areas under GENERAL and QUALIFICATION. Annotate in the flight block as "INIT QUAL".

3.2.2. Requal Evaluation: Evaluate all areas under GENERAL and QUALIFICATION. Upon successful completion of the re-qualification evaluation, the AF Form 8 will indicate crew position as "MB" or "IB" and annotate in the flight block as "RQ QUAL". **(T-2).**

3.2.3. Periodic Evaluation: Evaluate all areas under GENERAL and QUALIFICATION.

3.2.4. Emergency Procedures Evaluation "(EPE)". The "EPE" evaluation will be conducted in the BOT and shall be accomplished in conjunction with all qualification evaluations. **(T-2).** Use a contractor-developed AMC/A3VK-approved scenario. All scenarios will be run as written. **(T-2).** The following malfunctions will be evaluated as a minimum:

3.2.4.1. A Boom Camera, Primary Display, Or Boom Overlay Malfunction. **(T-2).**

3.2.4.2. A Boom Malfunction. **(T-2).**

3.2.4.3. A Flight Control Stick and/Or Telescope Control Stick Malfunction. **(T-2).**

3.2.4.4. An AAR Signal System Malfunction. **(T-2).**

3.2.4.5. Tanker Breakaway. **(T-2).**

3.2.4.6. Area 15, Air Refueling (Centerline Drogue). **(T-2).**

3.2.4.7. A WCA (Warning Caution Advisory) Response. **(T-2).**

3.2.4.8. A Boom or CDS Hydraulic System Malfunction. **(T-2). Note:** If a breakaway procedure cannot be evaluated in the BOT, it must be evaluated in-flight or rescheduled to evaluate the area.

3.3. Mission Evaluations. A mission evaluation will be administered to boom operators prior to performing unsupervised cargo loading/unloading duties. An EPE will be completed and annotated for all mission evaluations. **(T-2).** All MSN evaluations may be conducted in the FuT using a contractor-developed AMC/A3VK-approved scenario and will include a weight and balance computation, aircraft loading checklist (including the mission departure steps), cargo onload, a passenger briefing, cargo offload, and aircraft offloading checklist. **(T-2).** In-flight passenger handling procedures may be verbally evaluated. A minimum of six cargo pallets shall be used for each MSN evaluation (FuT only). **(T-2).**

3.3.1. Initial: Evaluate all areas under GENERAL, MISSION, and Area 17, Weight and Balance. Annotate in the flight block as "INIT MSN". If INIT MSN evaluation is

accomplished prior to INIT QUAL evaluation, annotate crew position as "MB" on INIT MSN evaluation AF Form 8.

3.3.2. Requal: Evaluate all areas under GENERAL, MISSION, and Area 17, Weight and Balance. Upon successful completion of the requal mission evaluation, the AF Form 8 will indicate crew position as "MB" or "IB" and annotate in the flight block as "RQ MSN". **(T-2).**

3.3.3. Periodic: Evaluate all areas under GENERAL, MISSION, and Area 17, Weight and Balance.

3.4. Instructor Evaluations. The flight examiners will place particular emphasis on the examinee's ability to recognize student difficulties and provide timely and effective corrective action. **(T-2).** Flight examiners will exercise sound judgment to ensure the explanations/demonstrations do not distract the examinee's attention or disrupt mission objectives. **(T-2).**

3.4.1. Initial: The QUAL portion of the INIT INSTR evaluation will be conducted on an actual air refueling mission. **(T-2).** Conduct evaluations with the examinee instructing a student boom (preferred) or examiner acting as a student boom in accordance with AFI 11-202V2, paragraph 5.2.4.2.1. INIT INSTR evaluations will include all areas under GENERAL, QUALIFICATION, and INSTRUCTOR. **(T-2).** The INIT INSTR evaluation will re-establish an individual's QUAL eligibility period. **(T-2).** If an overall grade is qualification level 3 (Q-3) in areas under GENERAL and/or QUALIFICATION, the individual will be graded Q-3 for both QUAL and INIT INSTR evaluations. **(T-2).**

3.4.2. Requal: The QUAL portion of the RQ INSTR evaluation will be conducted on an actual air refueling mission. **(T-2).** Conduct evaluations with the examinee instructing a student boom (preferred) or examiner acting as a student boom in accordance with AFI 11-202V2, paragraph 5.2.4.2.1. Requal Instructor evaluations will include all areas under GENERAL, QUALIFICATION, and INSTRUCTOR. **(T-2).**

3.4.3. Periodic: Instructor Boom Operators will be evaluated on their ability to instruct on all periodic evaluations. **(T-2).** Evaluate all areas under GENERAL, QUALIFICATION, MISSION, and INSTRUCTOR.

3.5. Emergency Procedures Evaluation (EPE). Evaluate an aircrew member's knowledge of emergency procedures and systems knowledge during QUALIFICATION and MISSION evaluations. The EPE should cover a cross section of aircraft systems. Examinees should be able to demonstrate an understanding of aircraft systems in emergency scenarios.

3.6. Boom Operator Grading Criteria.

3.6.1. **General.**

3.6.1.1. **Area 1, Directives and Publications.**

3.6.1.2. **Q.** Possessed a high level of knowledge of all applicable aircraft publications and procedures and understood how to apply this knowledge to enhance mission accomplishment. Publications were current and properly posted. Electronic Flight Bag was in proper configuration in accordance with MAJCOM directives.

3.6.1.3. **Q-.** Unsure of some directives but could locate information in appropriate publications. Publications were current but improperly posted. Electronic Flight Bag configuration was not in accordance with MAJCOM directives.

3.6.1.4. **U.** Unaware of established procedures and/or could not locate them in the appropriate publication in a timely manner. Publications were not current. Electronic Flight Bag configuration was not in accordance with MAJCOM directives and/or data was not current.

3.6.2. **Area 2, Mission Preparation/Planning.**

3.6.2.1. **Q.** Read and initialed for all items in FCIF. Completed/obtained all applicable forms. Complied with all local directives. Attended all required briefings.

3.6.2.2. **Q-.** Same as above except for minor deviations or omissions, which did not impair mission effectiveness. Did not fully comply with local directives, but did not detract from safety.

3.6.2.3. **U.** FCIF was not reviewed,initialed, or applied. Failed to attend required briefings. Failed to obtain/complete all applicable forms, or made major errors or omissions. Did not obtain adequate mission information. Failed to comply with local directives.

3.6.3. **Area 3, Use of Checklist.**

3.6.3.1. **Q.** Procedures and checklist items required by flight manual and applicable directives were accomplished in a thorough and proficient manner.

3.6.3.2. **Q-.** Procedures and checklist items required by flight manuals and applicable directives were accomplished with omission, deviation, or error, which detracted from the overall efficient conduct of the mission. Performance was the minimum acceptable.

3.6.3.3. **U.** Procedures or checklist items required by flight manual and applicable directives were accomplished with omission, deviation, or error which did, or could adversely affect the successful accomplishment of the mission or task.

3.6.4. **Area 4, Safety Consciousness (Critical).**

3.6.4.1. **Q.** Aware of and complied with all safety factors required for safe aircraft operation and mission accomplishment.

3.6.4.2. **U.** Not aware of or did not comply with all safety factors required for safe aircraft operation or mission accomplishment. Operated aircraft or equipment in a dangerous manner.

3.6.5. **Area 5, Judgment/Compliance (Critical).**

3.6.5.1. **Q.** Prepared and completed mission in compliance with existing regulations and directives. Demonstrated knowledge of operating procedures and restrictions.

3.6.5.2. **U.** Unaware of established procedures and/or could not locate them in the appropriate publication in a timely manner. Failed to comply with a procedure that could have jeopardized safety or mission success.

3.6.6. **Area 6, Crew Coordination/Crew Resource Management (CRM)/Threat and Error Management.** Use AFI 11-290_AMCSUP, applicable MAJCOM supplements, and AMC Form 4031, as a reference.

3.6.6.1. **Q.** Proactively applied appropriate, established CRM/TEM concepts throughout the flight/mission. Ensured safe/effective mission accomplishment by anticipating, recognizing, and mitigating relevant threats. Identified and mitigated own and other crewmembers' errors via the proper use of monitoring/crosschecking procedures.

3.6.6.2. **Q-.** Reactively and inconsistently, or inadequately applied appropriate, established CRM/TEM concepts, but did not allow those deficiencies to detract from mission accomplishment and/or flight safety. Unreliably and/or inadequately anticipated, identified, or mitigated relevant threats and/or own or other crewmembers' inconsequential errors.

3.6.6.3. **U.** Did not apply appropriate, established CRM/TEM concepts to ensure safe/effective mission accomplishment. Failed to anticipate, identify, or mitigate relevant threats and/or own or other crewmembers' consequential errors.

3.6.7. **Area 7, Communication Procedures.**

3.6.7.1. **Q.** Displayed a satisfactory knowledge of, and compliance with, correct communication procedures. Transmissions were concise and used proper terminology. Accomplished required calls and acknowledged transmissions in a manner which enhanced mission effectiveness.

3.6.7.2. **Q-.** Displayed adequate communication procedures, but was slow or not concise in making transmissions. Transmissions contained erroneous information or included non-standard terminology. Mission effectiveness was not jeopardized.

3.6.7.3. **U.** Incorrect procedures or poor performance caused confusion and jeopardized mission accomplishment. Omitted required transmissions or transmitted erroneous information.

3.6.8. **Area 8, Aircrew Flight Equipment/Egress.**

3.6.8.1. **Q.** Displayed thorough knowledge of location and use of aircrew flight equipment systems. Demonstrated and emphasized the proper operating procedures used to operate aircraft egress devices such as doors, windows, hatches, slide rafts, and escape ropes.

3.6.8.2. **Q-.** Limited knowledge of location and use of aircrew flight equipment systems. Unsure of the proper operating procedures used to operate some of the aircraft egress devices.

3.6.8.3. **U.** Displayed unsatisfactory knowledge of location and use of aircrew flight equipment systems. Unable to properly operate aircraft egress devices.

3.6.9. **Area 9, Knowledge/Completion of Forms.**

3.6.9.1. **Q.** All required forms were complete, accurate, readable, accomplished on time and in accordance with applicable directives/procedures in Aircrew e-Pubs and FAA/ICAO directives. Related an accurate description of significant events to applicable agencies (Safety, Maintenance, etc.).

3.6.9.2. **Q-.** Minor errors on forms that did not affect conduct of the mission. Incorrectly or incompletely reported some information due to minor errors, omissions, and/or deviations.

3.6.9.3. **U.** Did not accomplish required forms. Omitted or incorrectly reported significant information due to major errors, omissions, and/or deviations.

3.6.10. **Area 10, Airmanship/Situational Awareness (Critical).**

3.6.10.1. **Q.** Executed the assigned mission in a timely, efficient manner. Demonstrated strict professional flight and crew discipline throughout all phases of flight. Conducted the flight with a sense of understanding and comprehension.

3.6.10.2. **U.** Decisions or lack thereof, resulted in failure to accomplish the assigned mission. Failed to exhibit strict flight and crew discipline.

3.6.11. **Area 11, Ground Operations.** This area includes all ground activity through BEFORE TAKEOFF checklist.

3.6.11.1. **Q.** Complied with established station, start engine, taxi, and takeoff times to assure thorough pre-flight, check of personal equipment, crew/passenger briefings, etc. Accurately determined readiness of aircraft for flight. Completed all systems pre-flight/post-flight inspections in accordance with flight manuals.

3.6.11.2. **Q-.** Same as above except for minor procedural deviations that did not detract from mission effectiveness.

3.6.11.3. **U.** Errors directly contributed to a late takeoff that degraded the mission. Failed to accurately determine readiness for flight. Failed to pre-flight/post-flight a critical component or could not conduct a satisfactory pre-flight/post-flight inspection.

3.6.12. **Area 12, Systems Operations/Knowledge/Limitations.**

3.6.12.1. **Q.** Demonstrated/explained a complete knowledge of aircraft systems operations/limitations and proper procedural use of systems with minimal reference to flight manual/available aids.

3.6.12.2. **Q-.** Marginal knowledge of aircraft systems operations and limitations in some areas. Used individual technique instead of established procedure. Required moderate references to flight manual/available aids to differentiate between procedure and technique.

3.6.12.3. **U.** Unsatisfactory systems knowledge. Unable to demonstrate/explain the procedures for aircraft system operations.

3.6.13. **Area 13, Non-normal/Emergency Procedures (If Observed).** Boom operators will be graded on their initial response and actions taken to any actual emergency/abnormal conditions that occurs either in-flight or on the ground during the evaluation. **(T-2).**

3.6.13.1. **Q.** Operated within prescribed limits and correctly diagnosed problems. Performed/explained proper corrective action for each type of malfunction or abnormal condition. Effectively used available aids.

3.6.13.2. **Q-.** Operated within prescribed limits but slow to analyze problems or apply proper corrective actions. Did not effectively use checklist and/or available aids.

3.6.13.3. **U.** Exceeded limitations. Unable or failed to analyze problem or take proper corrective action. Did not use checklist and/or available aids.

3.7. QUALIFICATION

3.7.1. **Area 14, Air Refueling (Boom).**

3.7.1.1. **Q.** Demonstrated a satisfactory knowledge of procedures and equipment. Complied with directives. Coordinated with tanker and receiver pilots. Boom control was smooth and contacts were effective. Monitored receiver closely and gave corrections as necessary. Used proper procedures and techniques that would not jeopardize mission or safety.

3.7.1.2. **Q-.** Same as above except for minor deviations which did not or would not jeopardize safety or mission effectiveness. Boom control was slightly erratic resulting in contacts being delayed.

3.7.1.3. **U.** Failed to accomplish required checks. Boom control was erratic, and/or technique used in attempting contacts resulted in delays to such extent that fuel could not be offloaded within the time available. Inadequate knowledge, procedures, or techniques jeopardized safety of flight.

3.7.2. **Area 15, Air Refueling (Centerline Drogue System - CDS) (If Observed).**

3.7.2.1. **Q.** Demonstrated a satisfactory knowledge of procedures and equipment. Complied with directives. Coordinated with tanker and receiver pilots. Monitored receiver closely and gave corrections as necessary. Used proper procedures and techniques that would not jeopardize mission or safety.

3.7.2.2. **Q-.** Same as above except for minor deviations, which did not or would not jeopardize safety or mission effectiveness.

3.7.2.3. **U.** Failed to accomplish required checks. Inadequate knowledge, procedures, or techniques jeopardized safety of flight.

3.7.3. **Area 16, Air Refueling (Wing AAR Pods WARP) (If Observed).**

3.7.3.1. **Q.** Demonstrated a satisfactory knowledge of procedures and equipment. Complied with directives. Coordinated with tanker and receiver pilots. Monitored receiver closely and gave corrections as necessary. Used proper procedures and techniques that would not jeopardize mission or safety.

3.7.3.2. **Q-.** Same as above except for minor deviations which did not or would not jeopardize safety or mission effectiveness.

3.7.3.3. **U.** Failed to accomplish required checks. Inadequate knowledge, procedures, or techniques jeopardized safety of flight.

3.7.4. **Area 17, Weight and Balance (DD Form 365-4).** Number of errors will be considered even if no tolerances have been exceeded.

3.7.4.1. **Q.** Criteria.

3.7.4.1.1. Weight: Within 3000 lbs of actual aircraft gross weight.

3.7.4.1.2. CG: Error not in excess of 1 percent MAC.

3.7.4.2. **Q-.** Criteria.

3.7.4.2.1. Weight: Error exceeded 3000 lbs, but less than 5000 lbs.

3.7.4.2.2. CG: Error exceeded 1 percent, but less than 1.5 percent MAC.

3.7.4.3. **U.** Criteria. Exceeded Q- tolerances.

3.7.5. **Area 18, EPE.**

3.7.5.1. **Q.** Consistently used the correct checklist. Performed proper corrective action for each type of malfunction or abnormal condition. Effectively coordinated with other crewmembers throughout mission.

3.7.5.2. **Q-.** Checklist responses were untimely, with omission, deviation, or error which detracted from overall efficient conduct of the mission. Crew coordination was adequate to accomplish the mission. Performance was the minimum acceptable.

3.7.5.3. **U.** Used incorrect checklist. Unable to identify the correct checklist to use in a given situation. Procedures or checklist items were accomplished with omission, deviation, or error, which did, or could adversely affect the successful accomplishment of the mission or task. Demonstrated poor crew coordination, which negatively affected mission accomplishment.

3.8. MISSION.

3.8.1. **Area 19, Cargo Loading/Unloading.**

3.8.1.1. **Q.** Demonstrated a thorough knowledge of required procedures as outlined in the flight manual and applicable directives. Load planning was accomplished without errors or omissions. Required briefings were clear, concise and accurate. Coordinated with air terminal operation personnel (or equivalent) on cargo loading/unloading matters.

3.8.1.2. **Q-.** Demonstrated a limited knowledge of required procedures as outlined in the flight manual and applicable directives. Procedures were accomplished with errors or deviations which did not or would not detract from the cargo loading/unloading operation or mission. Load planning contained minor errors or omissions without exceeding established limits. Required briefings contained minor errors or omissions.

3.8.1.3. **U.** Demonstrated an unsatisfactory knowledge of required procedures as outlined in the flight manuals and applicable directives. Procedures were not complied with, which jeopardized mission accomplishment or the safety of the cargo loading/unloading operation. Required briefings were unclear and/or ineffective causing confusion. Failed to coordinate with air terminal operation personnel (or equivalent) on cargo loading/unloading matters. Load planning contained major errors or omissions and/or exceeded established limits.

3.8.2. **Area 20, Passenger Handling (If Observed, Else Verbally Evaluate).**

3.8.2.1. **Q.** Demonstrated a thorough knowledge of required passenger handling, normal/emergency procedures, and equipment as outlined in applicable guidance. Correctly briefed and performed passenger handling procedures.

3.8.2.2. **Q-.** Demonstrated a limited knowledge of required passenger handling, and related emergency procedures and equipment as outlined in applicable guidance. Difficulty briefing and/or performing passenger handling procedures.

3.8.2.3. **U.** Demonstrated an unsatisfactory knowledge of required passenger handling or related emergency procedures and equipment as outlined in applicable guidance. Failed to brief and/or did not perform proper passenger handling procedures that could jeopardize passanger safety/control.

3.8.3. **Area 21. NVG Usage/Limitations (If Observed).**

3.8.3.1. **Q.** Demonstrated or explained a thorough knowledge of NVG pre-flight, operations, limitations, and procedures.

3.8.3.2. **Q-.** Exhibited minimal knowledge of NVG pre-flight, operations, limitations, and procedures. Lack of knowledge did not impact mission effectiveness.

3.8.3.3. **U.** Failed to demonstrate or explain adequate knowledge of NVG pre-flight, operations, limitations, and procedures. Safety was compromised. Lack of knowledge impacted mission effectiveness.

3.9. Instructor.

3.9.1. **Area 22, Instructor Ability (Critical).**

3.9.1.1. **Q.** Demonstrated the ability to communicate effectively. Provided appropriate guidance when necessary. Planned ahead and made timely decisions. Identified and corrected potentially unsafe maneuvers/situations.

3.9.1.2. **U.** Unable to effectively communicate or provide timely feedback to the student. Did not provide corrective action when necessary. Did not plan ahead or anticipate student problems. Did not identify unsafe maneuvers/situations in a timely manner. Made no attempt to instruct.

3.9.2. **Area 23, Demonstration of Knowledge (Critical).**

3.9.2.1. **Q.** Effectively demonstrated procedures and techniques. Thorough knowledge of applicable aircraft systems, procedures, publications, and directives.

3.9.2.2. **U.** Did not demonstrate correct procedure or techniques. Insufficient depth of knowledge about applicable aircraft systems, procedures, and/or proper source material.

3.9.3. **Area 24, Student Briefing/Critique (Critical).**

3.9.3.1. **Q.** Briefings were well organized, accurate, and thorough. Reviewed student's present level of training and defined mission events to be performed. During the critique, demonstrated an effective ability to reconstruct the flight, offer mission analysis, and provide guidance where appropriate. Training grade reflected the actual performance of the student relative to the standard. Pre-briefed the student's next mission, if required.

3.9.3.2. **U.** Briefings were marginal or non-existent. Did not review student's past performance. Failed to adequately critique student or analyze the mission. Training grade

did not reflect actual performance of student. Overlooked or omitted major discrepancies. Incomplete pre-briefing of the student's next mission, if required.

JOSEPH T. GUASTELLA, Jr., Lt Gen, USAF
Deputy Chief of Staff, Operations

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 11-200, *Aircrew Training, Standardization/Evaluation, and General Operations Structure*, 21 September 2018

AFI 11-202V2, *Aircrew Standardization and Evaluation Program*, 6 December 2018

AFI 11-202V2_AMCSUP, *Aircrew Standardization and Evaluation Program,* 2 November 2020

AFI 11-215, *Flight Manual Program*, 25 March 2019

AFI 11-290_AMC Sup, *Cockpit/Crew Resource Management Program*, 4 December 2014

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFMAN 11-202V1, *Aircrew Training*, 27 September 2019

AFMAN 11-202V3, *Flight Operations*, 10 June 2020

AFMAN 11-2KC-46V1*, KC-46 Aircrew Training, 12 July 2021*

AFMAN 11-2KC-46V3*, KC-46 Operations Procedures, 12 July 2021*

AFMAN 11-210, *Instrument Refresher Program (IRP),* 4 October 2019

AFMAN 11-218, *Aircraft Operations and Movement on the Ground*, 5 April 2019

AFPD 11-2, *Aircrew Operations*, 31 January 2019

AFTTP 3-3.KC-46, *Combat Aircraft Fundamentals, KC-46,* 8 November 2019

ATP 3.3.4.2., *Allied Tactical Publication*

DAFI 33-360, *Publications and Forms Management*

Adopted Forms

AF Form 8, *Certificate of Aircrew Qualification*

AF Form 673, *Air Force Publication/Form Action Request*

AF Form 942, *Record of Evaluation*

AF Form 847, *Recommendation for Change of Publication*

AF Form 3862, *Flight Evaluation Worksheet*

AMC Form 4031, *CRM/TEM Skills Criteria Training/Evaluation*

DD Form 365-4, *Weight and Balance Clearance Form*

Abbreviations and Acronyms

AAR—Air to Air Refueling

AC—Aircraft Commander

ACIQ—Aircraft Commander Initial Qualification

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFPD—Air Force Policy Directive

AFTTP—Air Force Tactics Techniques Procedures

AMC—Air Mobility Command

ANG—Air National Guard

AOC—Air Operations Center

AR—Air Refueling

ARMS—Aviation Resource Management System

ASR—Airport Surveillance Radar

ATD—Aircrew Training Device

ATP—Allied Tactical Publication

BOT—Boom Operator Trainer

CCMD—Combatant Command

CDS—Centerline Drogue System

CG—Center of Gravity

COMAFFOR—Commander Air Force Forces

CONEMP—Concept of Employment

CRM—Crew Resource Management

DME—Distance Measuring Equipment

DOC—Designated Operational Capability

DTD—Data Transfer Device

EFB—Electronic Flight Bag

EFC—Expect Further Clearence

EMCON—Emission Control

EPE—Emergency Procedures Evaluation

ETOPS—Extended Operations

FCB—Flight Crew Bulletin

FCIF—Flight Crew Information File

FCOM—Flight Crew Operations Manual

FEF—Flight Evaluation Folder

FLIP—Flight Information Publication

FMC—Flight Management Computer

FPM—Flight Path Management

FuT—Fuselage Trainer

GA—Go Around

GBAS—Ground Based Augmentation System

GLS—GBAS Landing System

GPS—Global Positioning System

HHQ—Higher Head Quarter

IAN—Integrated Approach Navigation

IFF—Interigation Friend or Foe

ILS—Intrument Landing System

INIT—Initial

INST—Instrument

INSTM—Instrument

INSTR—Instructor

IP—Instructor Pilot

IRC—Instrument Refresher Course

JA/ATT—Joint Airborne/Air Transportability Training

KIAS—Knots Indicated Air Speed

LOC—Localizer

MAC—Mean Aerodynamic Chord

MAJCOM—Major Command

MAP—Missed Approach Point

MDA—Minimum Descent Altitude

MDN—Military Data Network

MDS—Mission Design Series

MP—Mission Pilot

MSN—Mission

NAF—Numbered Air Force

NAVAID—Navigational Aid

NDB—Non-directional Beacon

NM—Nautical Mile

NVG—Night Vision Goggles

OG/CC—Operations Group Commander

OGV—Operations Group Standardization/Evaluation

OI—Operational Instruction

OME—Operational Mission Evaluation

OPR—Office of Primary Responsibility

PA—Public Affairs

PAR—Precision Approach Radar

PCO—Pilot Check-out

PF—Pilot Flying

PFD/ND—Primary Flight Display/Navigation Display

PIC—Pilot In Charge

PIQ—Pilot Initial Qualification

PM—Pilot Monitoring

QRH—Quick Reaction Handbook

QUAL—Qualification

RNAV—Area Navigation

RNP—Required Navigation Performance

RQ—Re-qualification

SID—Standard Instrument Departure Route

SPINS—Special Instructions

SQB—Secure Question Bank

SQ/CC—Squadron Commander

SQ/DO—Squadron Director of Operations

STAN/EVAL—Standardization and Evaluation

STAR—Standard Arrival Route

TACAN—Tactical Air Navigation System

TACC—Tanker Airlift Control Center

TEM—Threat Error Management

TSAS—Threat Situational Awaraness System

TTP—Tactics Techniques Procedures

VFR—Visual Flight Rules

VOR—Very High-Frequency (VHF) Omnidirectional Range

VVM—Verbalize, verify, monitor

WARP—Wing Air Refueling Pods

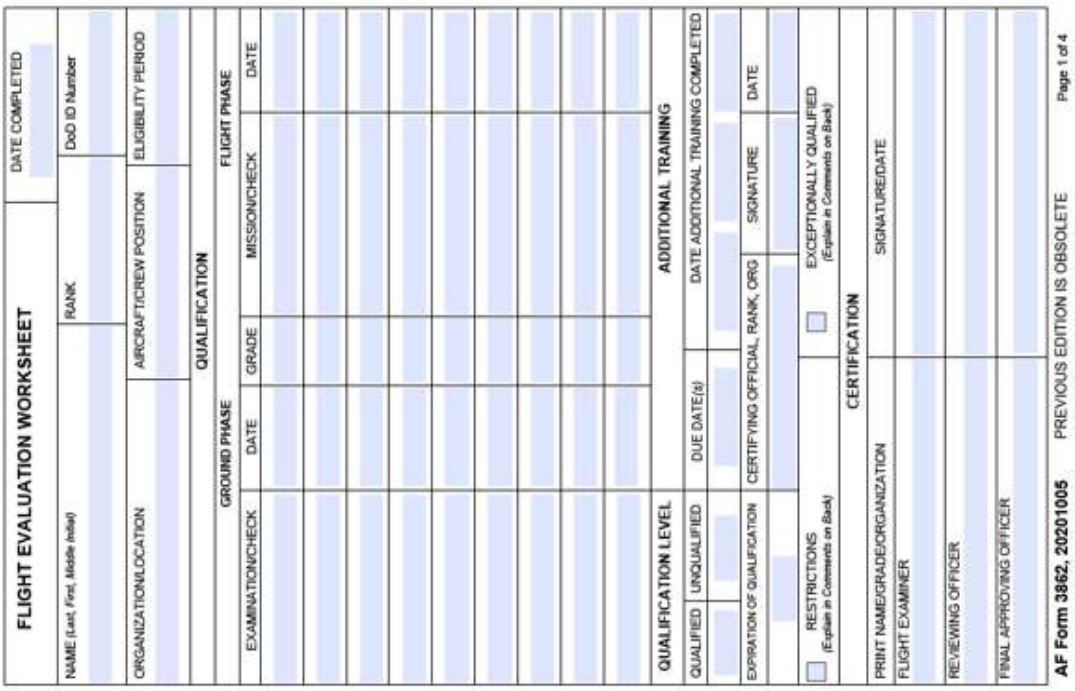
WCA—Warning Caution Advisory

WST—Weapon System Trainer

Attachment 2

FLIGHT EVALUATION WORKSHEET EXAMPLES

Figure A2.1. AF FORM 3862 (PAGES 1 & 4).



FLIGHT EVALUATION WORKSHEET

DATE COMPLETED

NAME (Last, First, Middle Initial) | RANK | DoD ID Number

ORGANIZATION/LOCATION | AIRCRAFT/CREW POSITION | ELIGIBILITY PERIOD

QUALIFICATION

GROUND PHASE

EXAMINATION/CHECK	DATE

FLIGHT PHASE

GRADE	MISSION/CHECK	DATE

QUALIFICATION LEVEL

QUALIFIED	UNQUALIFIED	DUE DATE(s)

ADDITIONAL TRAINING

DATE ADDITIONAL TRAINING COMPLETED

EXPIRATION OF QUALIFICATION | CERTIFYING OFFICIAL, RANK, ORG | SIGNATURE | DATE

☐ RESTRICTIONS (Explain in Comments on Back)

☐ EXCEPTIONALLY QUALIFIED (Explain in Comments on Back)

CERTIFICATION

PRINT NAME/GRADE/ORGANIZATION | SIGNATURE/DATE

FLIGHT EXAMINER

REVIEWING OFFICER

FINAL APPROVING OFFICER

AF Form 3862, 20201005 PREVIOUS EDITION IS OBSOLETE Page 1 of 4



AF Form 3862, 20201005 Page 4 of 4

Figure A2.2. KC-46 PILOT FLIGHT EVALUATION WORKSHEET (PAGES 1 & 4; 2 & 3).

KC-46 Pilot Evaluation	DATE COMPLETED: (Day/Month/Year)		
NAME (Last, First, MI), RANK		GRADE	SSN
		FROM GTIMS	GTIMS GENERATED
ORGANIZATION/LOCATION	ACFT/CREW POS KC-46A / IP/MP/FTP	ELIGIBILITY PERIOD GTIMS GENERATED	

QUALIFICATION			
GROUND PHASE		FLIGHT PHASE	
EXAM/CHECK	DATE	GRADE	MISSION/CHECK	DATE
Open Book			MSN	
Closed Book			INSTM/QUAL	
Memory Items			NO-NOTICE	
EPE			INIT INSTR / PILOT	
Instrument			REQUAL	

QUALIFICATION LEVEL		ADDITIONAL TRAINING			
QUALIFIED	INSQUALIFIED	DUE DATE(S)	DATE ADDITIONAL TRAINING COMPLETED		
1 2 3/1	3				
EXPIRATION DATE		CERTIFYING OFFICIAL, BASE AND O/G	SIGNATURE	DATE	
☐ RESTRICTIONS (Explain in Comments on Back)		☐ EXCEPTIONALLY QUALIFIED (Explain in Comments on Back)			

CERTIFICATION	
PRINT NAME/GRADE/ORGANIZATION	SIGNATURE/DATE
FLIGHT EXAMINER	
REVIEWING OFFICER	
FINAL APPROVING OFFICER	

AF IMT 3862 (AMC/A3V Approved 20 Feb 2020) Page 1 of 4

This evaluation was conducted IAW AFMAN 11-2KC-46V2 and (partially satisfies/satisfies) the requirements of an (SPOT/INST) periodic RQ/ (INSTR/MP/FTP)/(INSTM/QUAL/MSN/EPE) evaluation for the KC-46A conducted in the KC-46A (wingman system trainer/aircraft). All areas of this evaluation were completed in a (n) (satisfactory/unsatisfactory) manner. (Call Lt Col Maj/Capt Lt _____ during _____.)

For instructor evals: what specifically was instructed and when: (example instructed/ridden during mission planning)

This OME was conducted in conjunction with aircraft commander certification? (NO / YES)
Eval spans more than one day? (NO / YES) If yes, what other days? _____
Crewmember Directed Eval? (NO / YES)
Eval requires multiple evaluators to complete? (NO / YES) If yes, who was the other evaluator?
Any areas verbally evaluated? (NO / YES) If yes, please describe areas verbally evaluated below:

Supervisor debriefed at debrief? (NO / YES) If yes, please record the name of the supervisor _____
(AFI11-2C-V2: Unit CC or a designated representative will attend debrief on all less than Q-1 evaluations.)
Is this a recheck or re-qualification? (NO / YES) If yes, please provide the reason for recheck or re-qualification below:

Any discrepancies? (NO / YES) If yes, please provide details below. Please annotate the graded area, graded area type, graded area number, graded area title, grade, (e.g. Q-93), short discription for the discrepancy, and annotation if discrepancy was debriefed.

Recommended additional training? (No / Yes) If yes, please describe below:

Additional Comments:
Was the MSN (or Qual) portion of the evaluation given outside of the eligibility period to re-align the check-ride date? (NO / YES)
Any commendable items? (NO / YES) If yes, provide grading area number, grading area title, and flight examiner's description of commendable circumstances below:

AF IMT 3862 (AMC/A3V Approved 20 Feb 2020) Page 4 of 4

KC-46 PILOT FLIGHT EVALUATION WORKSHEET

AREA / SUB AREA	Q	Q-	U	T	REMARKS
GENERAL					
1. Directives/Publications					
2. Mission Prep/Planning/Performance					
3. Use of Checklist/Procedures					
4. Safety Consciousness (Critical)					
5. Judgment/Compliance (Critical)					
6. Crew Coordination/CRM/Threat & Error Mgt (TEM)					
7. Communication Procedures					
8. Aircrew Flight Equipment Systems / Egress					
9. Knowledge/Completion of Forms					
10. Airmanship/Situational Awareness (Critical)					
11. Aircrew Discipline					
QUALIFICATION					
12. Ground Operations/Taxi					
13. Takeoff					
14. Radar Operations/Weather Avoidance/Windshear					
15. Fuel Conservation					
16. VFR Pattern					
17. Landing					
17A. Normal Landing					
17B. Engine Out					
17C. Touch and Go					
18. Landing Roll/Braking					
19. All Engine Go-Around (GA)					
20. Engine Out Operations					
21. Engine Out GA/Engine Fail Takeoff Continued					
**. Memory Item Procedures (Critical)					
23. Other Observed Non-Normal Procedures					
24. Systems Operations/Knowledge/Limitations					
INSTRUMENT					
25. Instrument Departure/SID					
26. En Route Navigation/FMC Operations					
27. Holding (if observed)					
28. Use of NAVAIDS					
29. Descent/Arrival					
30. Precision Approach					
30A. PAR Approach					
30B. ILS/GLS Approach					
30C. Category II/III ILS					
31. Non Precision Approach					
32. Circling Approach (if observed)					
33. Missed Approach					
34. Automation Management					
35. Pilot Monitoring					

AF IMT 3862, (AMC/A3V Approved 20 Feb 2020) Page 2 of 4

AREA / SUB AREA	Q	Q-	U	T	REMARKS
MISSION					
36. Formation (if observed)					
37. TTP Execution					
38. Tanker A/R					
39. Receiver A/R					
40. Night Vision Goggles (if observed)					
40A. NVG Unaided Limitations (if observed)					
40B. NVG Takeoff (if observed)					
40C. NVG Approach and Landing (if observed)					
41. Nuclear Plans/Alert Procedures (if observed)					
INSTRUCTOR					
42. Instructor Ability (Critical)					
42A. Demonstration of Maneuvers (Critical)					
42B Student Briefing/Critique (Critical)					
SPECIAL INTEREST ITEMS (SIIs)					

TREND ANALYSIS GUIDANCE (Note: Use the "T" column to record trend information)
Grade each area use the following criteria:
A – Q is awarded and the evaluator makes a positive comment
B – Q is awarded and no other comment is warranted
C – Q or Q- is awarded and the evaluator makes a negative comment
D – U is awarded. Negative comments are mandatory

AF IMT 3862, (AMC/A3V Approved 20 Feb 2020) Page 3 of 4

Figure A2.3. KC-46 BOOM OPERATOR EVALUATION WORKSHEET (PAGES 1&4; 2 & 3).

KC-46 Boom Operator Evaluation	DATE COMPLETED:		
NAME (Last, First, MI)	GRADE	SSN	
		GTIMS Generated	
ORGANIZATION/LOCATION	ACFT/CREW POS KC-46 IB	ELIGIBILITY PERIOD GTIMS Generated	

GROUND PHASE			FLIGHT PHASE		
EXAM/CHECK	DATE	GRADE	MISSION/CHECK	DATE
Open Book			QUAL	
Closed Book				
EPE		1 / 2 / 3		
Memory Items		Q / U		

QUALIFICATION LEVEL		ADDITIONAL TRAINING	
QUALIFIED	UNQUALIFIED	DUE DATE(S)	DATE ADDITIONAL TRAINING COMPLETED
1 2	3		
EXPIRATION DATE		CERTIFYING OFFICIAL, RANK AND ORG	SIGNATURE / DATE

☐ RESTRICTIONS (Explain in Comments on Back) ☐ EXCEPTIONALLY QUALIFIED (Explain in Comments on Back)

CERTIFICATION	
PRINT NAME/GRADE/ORGANIZATION	SIGNATURE/DATE
FLIGHT EXAMINER	
REVIEWING OFFICER	
FINAL APPROVING OFFICER	

AF IMT 3862 (AMC/A3V Approved 20 Feb 2020) Page 1 of 4

COMMENTS:

QUAL:

This QUAL evaluation consisted of (day/night) Boom aerial refueling with a _____ receiver. All evaluation areas required by AFMAN 11-2KC-46V2 were accomplished in a (satisfactory/excellent) manner. _____ provided instruction on _____.

EVALUATOR BRIEFING:

- Type of Evaluation and Vol 2 Requirements
- Actions/expectations during Emergency
- See anything dumb/dangerous/different.....use CRM
- Interphone/Oxygen expectations

AF IMT 3862 (AMC/A3V Approved 20 Feb 2020) Page 4 of 4

KC-46 BOOM OPERATOR EVALUATION WORKSHEET

AREA / SUB AREA	Q	Q-	U	T	REMARKS
GENERAL					
1. Directives / Publications					
2. Mission Preparation / Planning					
3. Use of Checklists					
4. **Safety Consciousness (Critical)**		■			
5. **Judgment / Compliance (Critical)**		■			
6. Crew Coordination / CRM / TEM					
7. Communication Procedures					
8. Aircrew Flight Equipment / Egress					
9. Knowledge / Completion of Forms					
10. **Airmanship / Situational Awareness**		■			
11. Ground Operations					
12. Sys Ops / Knowledge / Limitations					
13. Non-normal / Emer Procedures (if observed)					
QUALIFICATION					
14. Air Refueling (Boom)					
15. Air Refueling (CDS) (if observed)					
16. Air Refueling (WARP) (if observed)					
17. Weight and Balance					
18. EPE					
MISSION					
19 Cargo Loading / Unloading					
20. Pax Handling (if observed, else verbal)					
21. NVG Usage / Limitations (if observed)					
INSTRUCTOR					
22. **Instructor Ability (Critical)**		■			
23. **Demonstration of Knowledge (Critical)**					
24. **Student Briefing / Critique (Critical)**		■			
SPECIAL INTEREST ITEMS (SIIs)					
Adherence to Guidance/Aircrew Discipline					
Reducing KC-46 Takeoff Performance Errors					
EFB Setup					

TREND ANALYSIS GUIDANCE (Note: Use the "T" column to record trend information)
Grade each area use the following criteria:
A – Q is awarded and the evaluator makes a positive comment
B – Q is awarded and no other comment is warranted
C – Q or Q- is awarded and the evaluator makes a negative comment
D – U is awarded. Negative comments are mandatory

AREA / SUB AREA	Q	Q-	U	T	REMARKS

If less than Q-1 will be awarded, notify Sq Stan/Eval as soon as possible.

Weight and Balance Grading Criteria

Grade	Discrepancy
Q	
Weight	Within 3000 lbs of actual aircraft gross weight.
CG	Error not in excess of 1 percent MAC
Q-	
Weight	Error exceeded 3000 lbs, but less than 5000 lbs.
CG	Error exceeded 1 percent, but less than 1.5 percent MAC
U	
Weight	Exceeded Q- tolerances
CG	

NOTE: Consider total number of errors even if no tolerances were exceeded

KC-46 Boom Operator Evaluation	DATE COMPLETED:		
NAME (Last, First, MI)	GRADE	SSN:	
		GTIMS Generated	
ORGANIZATION/LOCATION	ACFT/CREW POS KC-46 IB	ELIGIBILITY PERIOD GTIMS Generated	

GROUND PHASE			FLIGHT PHASE	
EXAM/CHECK	DATE	GRADE	MISSION/CHECK	DATE
EPE		1 / 2 / 3	MSN	

QUALIFICATION LEVEL	ADDITIONAL TRAINING	
(QUALIFIED) (UNQUALIFIED)	DUE DATE(S)	DATE ADDITIONAL TRAINING COMPLETED
1 2 3		
EXPIRATION DATE	CERTIFYING OFFICIAL, RANK AND ORG	SIGNATURE / DATE
☐ RESTRICTIONS (Explain in Comments on Back)	☐ EXCEPTIONALLY QUALIFIED (Explain in Comments on Back)	

CERTIFICATION	
PRINT NAME/GRADE/ORGANIZATION	SIGNATURE/DATE
FLIGHT EXAMINER	
REVIEWING OFFICER	
FINAL APPROVING OFFICER	

AF IMT 3862 (AMC/A3V Approved 20 Feb 2020) Page 1 of 4

COMMENTS:

MSN:

This MSN Evaluation consisted of a 6 Pallet Cargo load conducted in the KC-46 Fuselage Trainer (FuT). All areas required by AFMAN 11-2KC-46v2 were completed in a (satisfactory/excellent) manner. _____ provided instruction on _____ during loading/unloading.

Passenger Handling was verbally evaluated.

EVALUATOR BRIEFING:

- Type of Evaluation and Vol 2 Requirements (to include Passenger Handling)
- See anything dumb/dangerous/different.....use CRM

AF IMT 3862 (AMC/A3V Approved 20 Feb 2020) Page 4 of 4

KC-46 BOOM OPERATOR EVALUATION WORKSHEET

AREA / SUB AREA	Q	Q-	U	T	REMARKS
GENERAL					
1. Directives / Publications					
2. Mission Preparation / Planning					
3. Use of Checklists					
4. **Safety Consciousness (Critical)**		■			
5. **Judgment / Compliance (Critical)**		■			
6. Crew Coordination / CRM / TEM					
7. Communication Procedures					
8. Aircrew Flight Equipment / Egress					
9. Knowledge / Completion of Forms					
10. **Airmanship / Situational Awareness**		■			
11. Ground Operations					
12. Sys Ops / Knowledge / Limitations					
13. Non-normal / Emer Procedures (if observed)					
QUALIFICATION					
14. Air Refueling (Boom)					
15. Air Refueling (CDS) (if observed)					
16. Air Refueling (WARP) (if observed)					
17. Weight and Balance					
18. EPE					
MISSION					
19 Cargo Loading / Unloading					
20. Pax Handling (if observed, else verbal)					
21. NVG Usage / Limitations (if observed)					
INSTRUCTOR					
22. **Instructor Ability (Critical)**		■			
23. **Demonstration of Knowledge (Critical)**					
24. **Student Briefing / Critique (Critical)**		■			
SPECIAL INTEREST ITEMS (SIIs)					
Adherence to Guidance/Aircrew Discipline					
Reducing KC-46 Takeoff Performance Errors					
EFB Setup					

TREND ANALYSIS GUIDANCE (Note: Use the "T" column to record trend information)
Grade each area use the following criteria:
A – Q is awarded and the evaluator makes a positive comment
B – Q is awarded and no other comment is warranted
C – Q or Q- is awarded and the evaluator makes a negative comment
D – U is awarded. Negative comments are mandatory

AREA / SUB AREA	Q	Q-	U	T	REMARKS

"If less than Q-1 will be awarded, notify Sq Stan/Eval as soon as possible."

Weight and Balance Grading Criteria

Grade	Discrepancy
Q	
Weight	Within 3000 lbs of actual aircraft gross weight.
CG	Error not in excess of 1 percent MAC
Q-	
Weight	Error exceeded 3000 lbs, but less than 5000 lbs.
CG	Error exceeded 1 percent, but less than 1.5 percent MAC
U	
Weight	Exceeded Q- tolerances
CG	

NOTE: Consider total number of errors even if no tolerances were exceeded